

Comprehensive diffusion MRI analysis reveals key microstructural alterations underlying white matter hyperintensities

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Abstract: Cerebral small vessel disease (CSVD) is a prevalent age-related condition and a major contributor to cognitive decline, dementia, and stroke. Its most common MRI marker, white matter hyperintensities (WMH), reflects heterogeneous microstructural alterations, including demyelination, inflammation, and axonal loss. Monitoring of individual and regional variability associated with WMH in vivo provides critical insights into pathophysiological processes associated with aging and CSVD. However, diffusion MRI studies of WMH remain limited by the use of dichotomous lesion-based analyses, insufficient correction for free-water contamination, and an overreliance on fractional anisotropy. We analyzed longitudinal ADNI3 data from 216 cognitively healthy older adults with WMH over a two-year period using three complementary approaches: (1) lesion-gradient-based analyses to characterize microstructural alterations within WMH and the surrounding normal-appearing white matter; (2) an

aging-adapted tractography pipeline to examine diffusion profiles along major white matter bundles using tractometry; and (3) data-mining approaches applied to bundle profiles to identify the diffusion metrics most strongly associated with WMH burden. Lesion-gradient-based analyses revealed pronounced microstructural gradients extending from WMH into the surrounding normal-appearing white matter, which remained largely stable over time. Tractometry and data-mining analyses showed that the free-water fraction as well as tissue radial and mean diffusivities were strongly associated with WMH volume. Together, these findings indicate that WMH-related pathology extends beyond visibly affected tissue, with a progressive accumulation of free water within the tissue over time, accompanied by subtle alterations in tissue microstructure consistent with mild myelin-related changes, without evidence of substantial axonal compromise. Furthermore, this study suggests that FA, even when corrected for free-water, is not the optimal metric for assessing WMH-related microstructural alterations.

1 Introduction

Cerebral small vessel disease (CSVD) represents a critical condition, particularly prevalent among older populations (Debette et al., 2019), and is strongly associated with a wide range of cognitive and neurological disorders. The disease involves pathological changes affecting the small blood vessels of the brain, particularly arterioles, venules, and capillaries. These vascular alterations can lead to vascular cognitive impairment (VCI), dementia, and stroke (Rensma et al., 2018). On magnetic resonance imaging (MRI), CSVD manifests through several core markers, including white matter hyperintensities (WMH), microbleeds, lacunae, and perivascular spaces (Zhao et al., 2021).

Among these CSVD markers, WMHs stand out as the most prevalent for characterizing disease burden and cognitive impact. They affect over 90% of individuals with CSVD older than 65 years (Zhang et al., 2025) and can be reliably detected using standard FLAIR imaging (Barkhof & Scheltens, 2002). WMHs are strongly associated with cognitive decline, increased stroke risk, mortality, and late-onset depression (Karvelas & Elahi, 2023). [In particular, WMH volume has been shown to be interrelated with local cerebral network efficiency and information processing speed, suggesting that WMH-related cognitive impairment is partly mediated by alterations in white matter network organization \(Vergoossen et al., 2021\).](#) Neuropathological studies indicate that these [WMHs](#) variably reflect white matter degeneration, including axonal loss, demyelination, and gliosis, with heterogeneous pathophysiological origin (McAleese et al., 2021). Furthermore, both the location and intensity of WMHs influence not only the severity but

also the profile of cognitive impairments (O’Sullivan, 2024; Petersen et al., 2024). Better characterization of the global impact of WMH burden on the microstructural integrity of the white matter (WM) and its longitudinal progression in vivo is essential for understanding the pathophysiological processes involved in CSVD.

In MRI, diffusion imaging is the modality most widely used to characterize microstructural changes in white matter. Diffusion MRI provides information about the behavior of water molecules within tissues, allowing inferences about certain aspects of microstructural architecture. Diffusion tensor imaging (DTI) metrics are commonly used for this purpose and include fractional anisotropy (FA), mean diffusivity (MD), radial diffusivity (RD), and axial diffusivity (AD). However, in studies involving older populations, these diffusion measures can be influenced by increased free water content infiltrating the tissue, which alters diffusion values (Bergamino et al., 2021). As a result, the interpretation of conventional DTI metrics becomes more complex and may compromise the accuracy of downstream structural analyses if these effects are not carefully accounted for. For this reason, applying free-water correction to remove the contribution of extracellular free water is essential when investigating white matter microstructure in aging populations (Berger et al., 2022; Chang et al., 2025; Duering et al., 2018; Mayer et al., 2022).

The microstructural changes associated with WMH can be investigated at different spatial scales using diffusion imaging. One approach relies on a purely lesion-based framework that does not take into account the underlying WM pathways, aiming to characterize tissue damage within WMHs and to contrast it with the surrounding normal-appearing white matter (NAWM). WMHs boundaries are defined by a chosen segmentation method and correspond to visible intensity abnormalities, most typically on FLAIR. However, tissue alterations induced by WMHs do not occur in a strictly dichotomous manner separating “healthy” from “lesioned” tissue (Silbert et al., 2024). Instead, a gradient of microstructural disruption surrounds the WMHs, forming what is often referred to as the WMH penumbra (Maillard et al., 2011). Despite its relevance, this penumbral region remains largely overlooked in WMH research (Mayer et al., 2022). Considering this transitional zone is crucial, as investigating progressive tissue changes around WMHs may provide a more comprehensive understanding of the mechanisms and extent of WMH-related microstructural changes.

A second type of approach focuses on major WM pathways and applies tractography, followed by either manual or automatic bundles segmentation. Manual dissection approaches, such as those described by Catani et al. (Catani et al., 2002), rely on expert-driven identification of bundles, whereas automatic methods include White Matter Query Language (Wassermann et al., 2016), XTract in FSL (Warrington et al., 2020), WMA (O’Donnell & Westin, 2007), and BundleSeg (St-Onge et al., 2023). However, tractography-based bundle identification alone does not precisely account for the spatial distribution of

WMHs along these bundles, making it difficult to fully capture the microstructural impact of WMHs (Kuang et al., 2023; Muñoz Maniega et al., 2019; Rheault et al., 2020; Rizvi et al., 2020). In order to allow a spatially more fine-grained analysis along the bundles, tractometry, or tract profiling (Colby et al., 2012; Yeatman et al., 2012) can be performed. These approaches subdivide fiber bundles into a predefined number of segments, enabling the analysis of diffusion metric variations along their trajectories. They allow the integration of WMH information, with spatial information along the trajectories of identified bundles (Chandio et al., 2020; Pichet Binette et al., 2021; Qiu et al., 2024; Rheault et al., 2017; Roy et al., 2020).

One of the major drawbacks of tractometry is its complexity, both in terms of implementation and result interpretation. Tractography-based approaches are inherently sensitive to methodological variability, an issue that is further amplified in tractometry. Its application, therefore, requires particular caution, especially during tractogram generation, where segmentation and registration constitute major sources of reproducibility errors (Jbabdi & Johansen-Berg, 2011; Jones et al., 2013; Rheault et al., 2020). The problem can be exacerbated in the presence of WMH, which likely does not represent complete tissue destructions (Chang et al., 2025) but impacts diffusion properties that tractography approaches depend on. It necessitates the use of specific processing parameters to ensure the accurate reconstruction of any preserved bundles (Pichet Binette et al., 2021; Rheault et al., 2017; Roy et al., 2020). From an interpretative standpoint, tractometry produces a large volume of data, which can complicate its usability. It is generally not feasible to display or analyze all available measurements. As a result, many studies report only a subset of findings, typically guided by statistical significance or prior literature when selecting the most “relevant” bundles or metrics. This practice can introduce potential biases, particularly in the study of WMHs, as it focuses on well-known metrics and on bundles showing the strongest effects, without systematically examining the behavior of the remaining metrics or pathways.

To address these issues, exploratory, data-driven methods such as data-mining algorithms offer a promising alternative. These approaches allow for a systematic exploration of relationships between diffusion metrics and the full set of white matter bundles, providing a detailed and unbiased characterization of microstructural alterations captured by tractometry. By performing such exploratory analyses prior to hypothesis-driven testing, we can identify the most relevant metrics associated with pathological changes, thereby reducing interpretative bias.

In this study, we propose a robust analytical framework based on three complementary approaches to comprehensively characterize the impact of WMH on microstructural alterations and its progression in the 216 cognitively normal adults of the ADNI3 cohort, with longitudinal diffusion MRI data two years apart (Fig.1). [First, we applied a lesion-gradient-based analysis to characterize microstructural alterations](#)

within WMHs and their surrounding penumbra (0–12 mm around the WMH). This approach aimed to capture the gradient of tissue disruption surrounding the WMH core and to investigate progressive microstructural changes in this transitional region. Second, we adapted a tractography pipeline specifically designed for aging populations (Pichet Binette et al., 2021; Qiu et al., 2024; Roy et al., 2020) to explicitly account for WMHs, enabling tractometry-based profiling of free-water–corrected diffusion metrics along major white matter bundles and capturing microstructural changes along anatomically defined pathways. Third, we applied data-mining algorithms to bundle-wise diffusion profiles to systematically explore associations between diffusion metric alterations and both WMH presence and WMH volume changes, with the aim of uncovering patterns that may not be detectable using conventional approaches. The overall goal of this study was to evaluate the global impact of WMH burden on white matter microstructural integrity and to examine how WMH characteristics relate to structural alterations within white matter pathways.

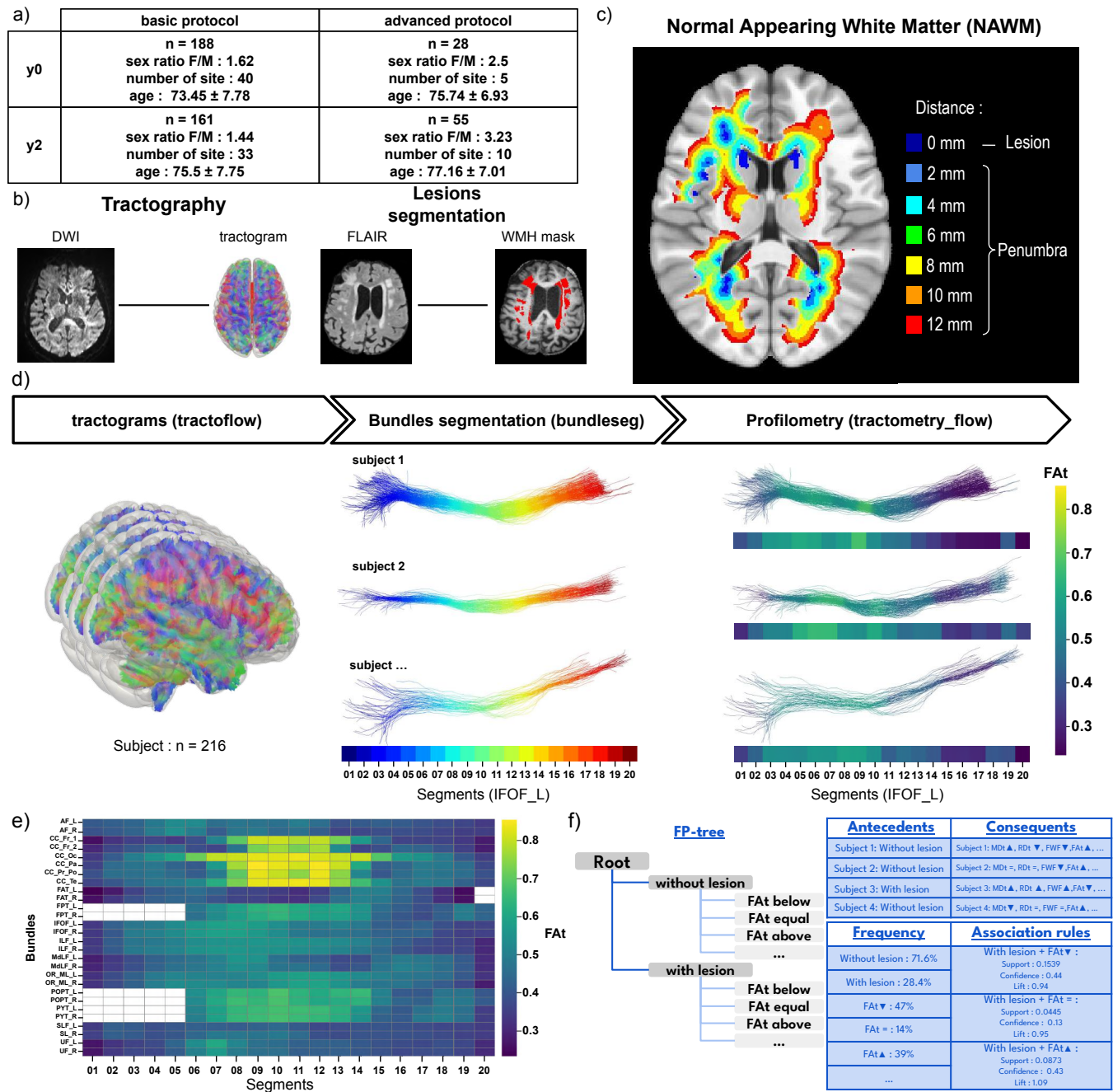


Figure 1: Summary of the study, illustrated with an example of the IFOF_L bundle profile computed using the FAt metric.

(a) Descriptive table of the ADNI3 cohort used for the study for the two acquisitions (y0 and y2). (b) Preprocessing of the raw data was performed, with the tractogram extracted from the DWI data via tractoflow and the WMH masks generated from the FLAIR images via SHIVA. (c) Normal-Appearing White Matter (NAWM). WMHs (0 mm) and perilesional penumbra (2–12 mm) were investigated using concentric 2 mm isotropic ROIs. Diffusion metric values were computed within each ROI. (d) Generation of bundle profiles using bundleseg and tractometry_flow for the various DTI metrics as well as lesion-related metrics. (e) tractometry : Population-level analysis of bundle profiles for the totality of bundles for one time point for FAt metrics. (f) Overview of the FP-Growth algorithm workflow. Each bundle segment is processed independently following the same procedure. First, an FP-tree is constructed, where WMH states are defined as antecedent (with and without) and diffusion metrics as consequent items (above Δ , below ∇ , or equal = relative to baseline). Based on the frequency of each item within the dataset, the FP-tree compactly represents recurring co-occurrence patterns. Association rules are then extracted for all possible antecedent–consequent pairs. For each pair, three key metrics are computed, including support (frequency of co-occurrence), confidence (conditional probability of the consequent given the antecedent), and lift (strength of association relative to independence).

2 Materials and methods

2.1 Study design and participants

Our study used data from the Alzheimer’s Disease Neuroimaging Initiative (ADNI), a multicenter project, with a specific focus on the third cohort (ADNI-3). ADNI-3 began in 2016 and includes participants recruited from 59 research centers across the United States and Canada, with annual data acquisitions. The study recruited men and women aged 55 to 90 years and categorized participants into three cohorts: cognitively normal (CN), mild cognitive impairment (MCI), and mild Alzheimer’s dementia. For the present study, we selected 216 participants from the CN cohort, without stratification by age or sex, each of whom underwent two longitudinal imaging sessions separated by two years (Fig.1.a).

2.2 MRI acquisition

Each participant in the ADNI protocol underwent scanning on a 3 T MRI scanner. Specific acquisition parameters were as follows: T1-weighted structural imaging was performed using an MP-RAGE sequence with TE = minimum full echo, TR = 2300 ms, TI = 900 ms, a field of view of $240 \times 256 \text{ mm}^2$, and isotropic voxel size of 1 mm^3 . Diffusion-weighted imaging (DWI) data were acquired using two protocol types that evolved over longitudinal acquisitions (Fig.1.a). The first, referred to as the basic protocol, used an echo-planar imaging (EPI) sequence with approximately 41 directions, TE = 56 ms, TR = 7200 ms, an acquisition matrix of $232 \times 232 \times 160 \text{ mm}^3$, an isotropic voxel size of 2 mm^3 , and a single shell at $b = 1000 \text{ s/mm}^2$. The second, referred to as the advanced protocol, maintained the same acquisition matrix and voxel size, but had TE = 71 ms, TR = 3300 ms, with multi-shell diffusion weighting ($b = 500, 1000, 2000 \text{ s/mm}^2$) and 112 directions. T2 FLAIR scans were acquired using an EPI sequence with TE = 119 ms, TR = 4800 ms, TI = 1650 ms, an acquisition matrix of $232 \times 232 \times 160 \text{ mm}^3$, and a voxel size of $1.2 \times 1 \times 1 \text{ mm}^3$. Due to the large number of acquisition sites and the variety of scanners used for ADNI data collection, acquisition protocols may vary across sites. For detailed site-specific acquisition information, please refer to the ADNI website.

2.3 WMH segmentation

We segmented WMHs using the SHIVAI preprocessing and deep learning segmentation workflow (Tsuchida et al., 2024). This workflow relies on a U-net convolutional neural network, adapted from the model proposed by (Boutinaud et al., 2021), to detect WMHs as markers of CSVD (Fig.1.b). Each WMH was then

labeled using two scripts from the scilpy library (Renauld et al., 2026). The first script assigned labels to WMHs within each individual, while the second harmonized WMH labels across the two acquisitions for the same individual, ensuring consistent WMH tracking over time.

2.4 Preprocessing and diffusion metrics extraction

2.4.1 TractoFlow

We computed diffusion metrics and generated all tractograms using the TractoFlow pipeline (Fig.1.b, Figure S1). TractoFlow (Theaud, Houde, et al., 2020) is an automated and reproducible diffusion MRI processing pipeline designed for large-scale tractography studies. It integrates diffusion MRI processing with T1-weighted image processing to obtain anatomical tissue segmentations used for tractography. Diffusion MRI preprocessing steps from the scilpy library (Renauld et al., 2026) include denoising, motion and distortion correction with FSL topup and eddy, bias field correction, and normalization. Preprocessed images were then fitted using constrained spherical deconvolution to enable local tractography, which was performed using local tracking models. TractoFlow emphasizes reproducibility, scalability, and compatibility with high-performance computing environments, thereby enabling standardized and reliable analyses of white matter pathways and their associated diffusion properties across subjects and sites.

Because our study focused on an elderly population, conventional tissue segmentation proved insufficient, as WMHs often lead to inaccurate tissue classification (Dadar et al., 2021; Oi et al., 2023). Altered image contrast within WMHs disrupts standard segmentation tools such as FAST (Figure S2). To address this limitation, we performed tissue segmentation using atlas-based methods (Gopinath et al., 2024; Roy et al., 2020; Theaud, Houde, et al., 2020) and incorporated WMH maps generated by SHiVAi into the seeding and tracking masks. These adjustments ensured comprehensive WMH inclusion and minimized bias introduced by white matter segmentation errors.

Additionally, the multicenter nature of the ADNI3 cohort required extensive harmonization across participants due to the large number of acquisition sites involved (59 sites in our study). Given this heterogeneity, no single ADNI site could serve as an unbiased internal reference for the others, and an external reference population was therefore required. We used the Cam-CAN cohort, the Cambridge Centre for Ageing and Neuroscience dataset, a large population-based sample ($n \approx 650$) spanning the adult lifespan, as this reference, as its size and homogeneity of acquisition provided a stable distribution against which the numerous ADNI sites could be aligned. We applied ComBat harmonization, following the approach originally developed for multi-site diffusion MRI data by Fortin et al. (2017) and further detailed by Jodoin et al. (2025), to mitigate variability related to acquisition site while preserving bio-

logically meaningful variability associated with age and sex. In a second step, to ensure comparability across the two acquisition protocols, all diffusion data were harmonized into a common single-shell framework: for participants scanned under the advanced multi-shell protocol, the $b = 1000 \text{ s/mm}^2$ shell was extracted from the multi-shell acquisition to match the structure of the standard single-shell protocol data, and FW-DTI metrics were then estimated from this harmonized single-shell dataset for all subjects regardless of original protocol. This shell-extraction step was validated in a subset of participants scanned under both protocols, by comparing metrics derived from their natively acquired single-shell data with those derived from the extracted $b = 1000 \text{ s/mm}^2$ shell of their multi-shell data, revealing no significant differences between the two. A detailed description of the processing pipeline, the validation analysis, and the specific parameter settings is provided in the supplementary materials (Supplementary Data S3 and S4) and in the official *TractoFlow* documentation. The pipeline is available on GitHub: <https://github.com/scilus/tractoflow>. Additionally, guidelines for best practices in DTI analysis of aging populations are provided in the supplementary materials (Figure S9).

2.4.2 Freewater metrics

To enhance WMH characterization, we employed `freewater_flow` (Di Tommaso et al., 2017; Kurtzer et al., 2017) to extract advanced diffusion metrics that extend beyond conventional tensor-based measures. The modeling of different signal components was performed using Accelerated Microstructure Imaging via Convex Optimization (AMICO), based on the framework proposed by (Daducci et al., 2015). This approach estimates and removes the contribution of extracellular free water from the diffusion signal, thereby enabling the computation of free-water-corrected diffusion metrics (FAt, MDt, RDt, ADt) that more accurately reflect tissue microstructural integrity. This correction is particularly valuable in lesion-prone regions, where increased extracellular water content can bias traditional diffusion metric estimates (Pasternak et al., 2009). In addition, `freewater_flow` computes the free water fraction (FWF), which quantifies the proportion of extracellular free water within each voxel. The modeling of these signal components is optimally performed using multi-shell acquisitions; however, it remains feasible with single-shell data, albeit with a coarser approximation of the different signal contributions (Chang et al., 2025) and see supplementary data (Figure S3 & S4). The pipeline is available on GitHub: https://github.com/scilus/freewater_flow.

2.5 Lesion-gradient-based analyses

2.5.1 WMH penumbra regions of interest (ROI) generation

Our first approach consisted of lesion-gradient-based analyses. For each subject and timepoint, we generated six layers of concentric rings around WMH masks, with each ring having a thickness of 2 mm (Fig.1.c), using a script from the *scipy* library (Renauld et al., 2026). The propagation was limited within the WM mask, creating the six NAWM layers that represented increasing distances from the WMH, up to 12 mm, and served as WMH penumbra regions of interests (ROIs). We excluded voxels overlapping with previously generated ROIs at each dilation step to prevent duplicate voxel inclusion across ROIs. We calculated the mean value of each diffusion metric within each ROI for every individual, [see supplementary data \(Figure S5 & table S1\)](#).

2.5.2 Statistical analyses

To assess spatial gradients of microstructural alterations within WMHs and the surrounding penumbra, their longitudinal progression, and the impact of WMH load on both gradients and progression, we applied separate linear mixed-effects models for each microstructural metric (FWF, FAt, MDt, ADt, and RDt). Each model included fixed effects for spatial distance (WMH and penumbra layers), time, and total WMH volume, and their interactions: Specifically, distance $\times$ time tested the longitudinal change in the microstructural gradients, distance $\times$ total WMH tested the impact of WMH load on the gradients, time $\times$ total WMH the impact of load on longitudinal progression. Finally, the three-way interaction tested the impact of WMH load on longitudinal change in the gradients. To normalize the distribution of WMH volumes, reduce the influence of extreme values, and improve model stability, total WMH volume was $\log(x+1)$ -transformed. All models were further adjusted for age and sex. Subject-specific random intercepts were included to account for within-individual correlations across layers and timepoints. Distance from the WMH boundary (0 to 12 mm) was modeled using a 6th-degree orthogonal polynomial ($\text{poly}(\text{distance}, 6)$) to allow sufficient flexibility in capturing non-linear spatial gradients of microstructural change.

2.6 Tractometry

2.6.1 Bundle extraction

The second approach was based on a WM bundle-based analysis (Fig.1.d). We used whole-brain tractograms generated with TractoFlow and segmented white matter bundles using BundleSeg (St-Onge et al., 2023). It provides a shape-based tractography segmentation method that leverages a white matter atlas to identify and extract specific bundles without relying on streamline clustering. Using a Fast Streamline Search approach (Garyfallidis et al., 2018), this method enables efficient and precise segmentation by directly comparing streamline shapes within a defined spatial radius. Previous work (Rheault, 2020) has demonstrated that this approach improves both speed and accuracy, particularly in regions with complex anatomy such as the brainstem and cerebellar tracts. The pipeline is available on GitHub: https://github.com/scilus/rbx_flow. In this study, we segmented whole-brain tractograms using an atlas of 39 bundles available on Zenodo (<https://zenodo.org/records/10103446>). We retained only 28 bundles, selecting those that were directly traversed by WMHs. The 28 bundles included: Six sub-divisions of Corpus Callosum (CC) (Frontal (Fr_1 & _2), Occipital (Oc), Parietal (Pa), Precentral-Postcentral (Pr.Po), Temporal (Te)); 11 pairs of association bundles in the left (L) and right (R) hemispheres, which were Arcuate Fasciculus (AF), Frontal Aslant Tract (FAT), Frontopontine Tract (FPT), Inferior Fronto-Occipital Fasciculus (IFOF), Inferior Longitudinal Fasciculus (ILF), Middle Longitudinal Fascicle (MdLF), Superior Longitudinal Fasciculus (SLF), and Uncinate Fasciculus (UF); and 4 pairs of projection bundles, including Optic Radiation (OR_ML), Parieto-Occipital Pontine Tract (POPT), Pyramidal Tract (PYT). The visual presentation of the different bundles is provided in the supplementary data (Figure S6, S7, S8).

2.6.2 Tractometry

Tractometry_flow (Cousineau et al., 2017; Di Tommaso et al., 2017; Kurtzer et al., 2017) was used to perform tractometry for each individual white matter bundle, projecting average metrics onto 20 equidistant sections along each bundle (Fig.1.d, Figures S5-S7). Tractometry_flow is a tool designed to segment white matter bundles and compute diffusion- and lesion-related metrics along their trajectories. It builds on the tool developed by (Cousineau et al., 2017). It generates a distance map, which measures the distance from each streamline to its nearest centroid point, and a label map, which divides each bundle centroid into a fixed number of equidistant sections and assigns this segmentation to all streamlines. Here we divided each bundle into 20 sections based on a common reference centroid across all subjects to generate bundle-wise profiles of diffusion metrics and segment-wise WMH volume proportion, both at

individual subject-levels and average profiles across subjects (Fig. 1e). Sections 1 to 5 of the FPT, POPT, and PYT bundles were excluded due to poor tracking at the pontine level. Similarly, section 20 of the FAT bundle was removed because it exhibited tracking defects in the majority of subjects. The pipeline is available on GitHub: https://github.com/scilus/tractometry_flow.

2.6.3 Statistical analyses

For each segment of the 28 bundles, we conducted correlation analyses independently for each diffusion metric, to test their association with the segment-specific WMH volume fractions, collapsing across the two timepoint (Fig.1.e) It allowed us to qualitatively assess whether WMH presence similarly affects different diffusion metrics. In order to assess the impact of WMH progression on the longitudinal change in microstructural properties, the same independent strategy was applied to test the correlation between segment-specific WMH volume fraction change with the change in each of the diffusion metrics. For each bundle and segment, we computed p-values and subsequently corrected them for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) procedure.

2.7 Data mining

In parallel with the segment-wise correlation analyses, we conducted a hypothesis-free, data-driven exploration of tractometry data with a data-mining framework based on the Frequent Pattern Growth (FP-Growth) algorithm (Fig.1.f). Data mining identifies latent structures or recurring associations within complex datasets without relying on predefined statistical models (Aggarwal, 2014; Han et al., 2000; Han et al., 2011), unlike conventional analyses, which typically test specific, theory-driven hypotheses and may therefore overlook unexpected relationships in high-dimensional settings (Fig.2.a).

While the method is inherently hypothesis-free, its application is constrained within the framework of two complementary analyses, each addressing a distinct purpose: one targeting cross-sectional patterns of WMH occurrence and its association with microstructural profiles, and the other focusing on longitudinal changes in WMH occurrence in relation to diffusion metrics. Each segment in each bundle and subject was first represented by a set of categorical descriptors (Fig.2.b) indicating WMH status (e.g., WMH presence/absence in a given segment for cross-sectional analysis, WMH increase/no change for longitudinal analysis) and diffusion metric deviations relative to a lesion-free baseline (e.g., MDt above baseline, FAt below baseline) or longitudinal change (e.g., MDt increase, FAt decrease). For cross-sectional analysis, we defined the lesion-free population-level reference for each diffusion metric in each segment by computing mean and standard deviations in subjects without WMH in the given segment. Individual

diffusion metrics were classified as “above” or “below” relative to the baseline standard deviation. Similarly, for longitudinal analysis, the population-level delta values from non-lesioned segments were used to define a segment-specific longitudinal baseline. Individual segment changes were then classified as increased, decreased, or unchanged relative to this baseline standard deviation.

With these categorical descriptors, the algorithm first constructs a compact tree structure (the FP-tree) which encodes how frequently individual descriptors and their combinations occur across subjects. Only patterns occurring at a minimum support threshold is retained, which was set to 5% of the cohort, corresponding to a minimum of 11 subjects in our study. In the second step, the frequent co-occurring patterns are used to generate association rules using FP-growth. In this framework, the antecedent represents a condition of interest (e.g., the presence of a WMH or a change in WMH volume), while the consequent represents a diffusion metric status (e.g., an increase in MDt).

Each association rule was evaluated with standard data-mining metrics, including support, confidence, and lift. Support quantifies the proportion of subjects in whom a given combined pattern occurs; confidence measures the conditional probability of observing the diffusion status given the WMH condition; and lift quantifies the strength of the association relative to chance. Confidence values range from 0 to 1, with values below 0.5 indicating low reliability, 0.5–0.7 moderate reliability, and above 0.7 high reliability of the rule. Lift values greater than 1 indicate a positive association, with 1.0–1.2 considered weak, 1.2–1.5 moderate, and 1.5–2.0 strong; values close to 1 indicate no association, while values below 1 indicate a negative association. In our large dataset, the most significant lift values ranged from 1.3–1.5. Although moderate in magnitude, these values represent consistent positive associations, indicating that co-occurrences occur slightly more often than expected by chance.

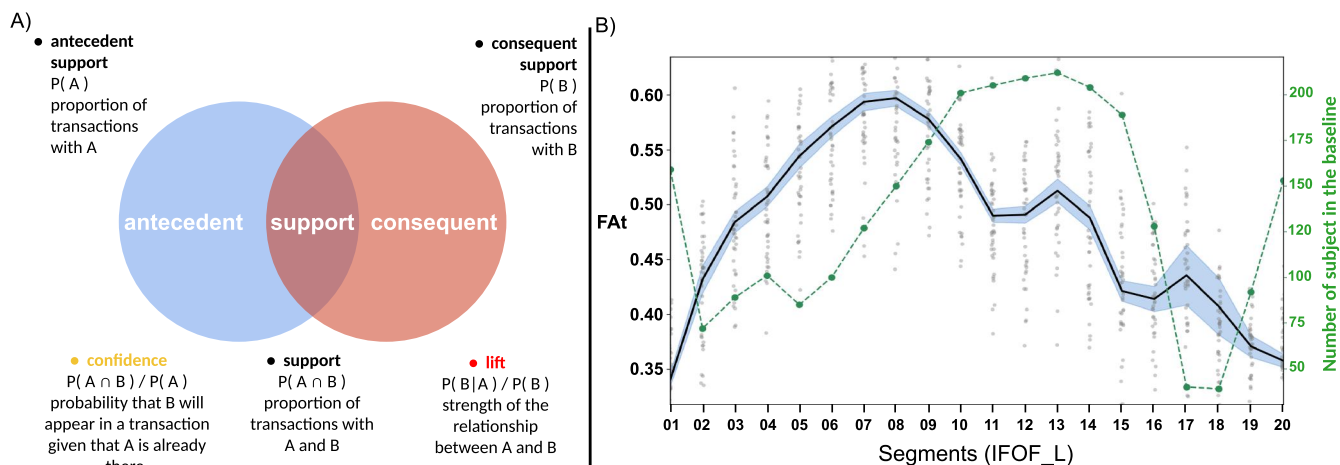


Figure 2: FP-Growth: Metrics and FP-Tree Classification

(a) Overview of the statistical procedures applied in the FP-Growth analysis. (b) Cross-sectional classification of subjects was performed based on their metric values relative to a healthy reference baseline (black line). This baseline was defined as the mean value computed from lesion-free segments (number shown as green line). When an individual's value (grey point) deviated significantly from the baseline, outside the standard deviation (blue band), it was classified as either increased or decreased, depending on the direction of the deviation. Values falling within the standard deviation of the baseline were classified as equal to the baseline. The same procedure was applied to longitudinal classification, using a baseline defined as the mean change observed in lesion-free segments. Subjects exhibiting a significant deviation from this standard deviation-based baseline were classified as showing either an increase or decrease over time. Values within the standard deviation were considered stable.

3 Results

3.1 lesion-gradient-based analyses

3.1.1 Spatial distribution and evolution of WMH

In this cognitively normal cohort the overall WMH burden was relatively low. The baseline WMH density map generated from the full cohort at y0 revealed that WMHs were most frequent in periventricular regions, particularly adjacent to the lateral ventricles (Fig.3.a), reaching the prevalence of WMH as high as 70% (red) in these regions. However, outside of the periventricular regions, WMH density tended to show lower values, with the majority of voxels showing 1% (blue), reflecting the heterogeneity of WMH spatial distributions across subjects. Longitudinally (Fig. 3.b), the total WMH volume showed a significant increase over the two-year period (paired t-test: $t = 9.899$, $p = 2.81e-19$), with the median log-transformed WMH volume rising from 8.307 (Y0) to 8.455 (Y2). Although 28 subjects (12.96%) exhibited a decrease in WMH volume, the majority of subjects (188, 87.04%) showed an increase. The relatively modest median change reflects limited overall progression, yet the positive t-test confirms a

statistically significant increase in WMH volume.

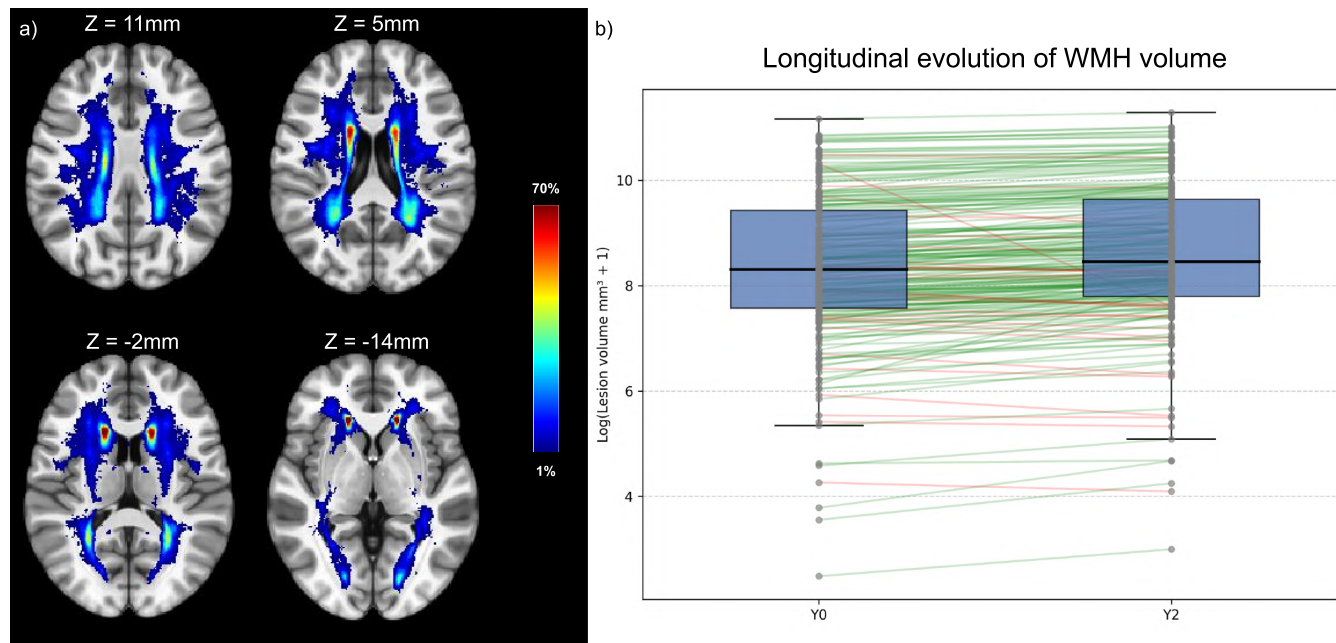


Figure 3: Density and evolution of WMH

(a) WMH were registered in MNI space for the 216 cognitively normal participants from ADNI3. Colors represent the proportion of subjects with a WMH in each voxel. (b) Longitudinal comparison of total WMH volume for each subject. Each subject is depicted by a gray point, with lines connecting their measurements across the two timepoints. Green lines denote an increase in WMH volume, whereas red lines indicate a decrease.

3.1.2 Load-dependent spatial gradients of microstructure in WMH and NAWM

Across all metrics, we observed clear spatial gradients of diffusion properties as a function of distance from WMHs, extending up to 12 mm from the WMH border (Fig.4). Both the free water fraction (FWF) and diffusivity metrics (ADt, MDt, RDt) were markedly elevated within WMHs and gradually decreased with increasing distance. Among the diffusivity metrics, ADt exhibited the most spatially extended abnormalities, showing a gradual decline across perilesional NAWM. FAt displayed a bell-shaped profile, with the lowest values within WMHs, increasing up to approximately 4 mm from the lesion, and then slowly decreasing at greater distances. Statistical analysis revealed no significant three-way interaction between spatial distance, timepoint, and global WMH burden, indicating that WMH load-dependent spatial gradients were stable longitudinally. Accordingly, the three-way interaction was removed, and the model focused on two-way interactions. A robust interaction between spatial distance and global WMH burden was observed for all metrics (all $p < 0.001$), indicating that the spatial profiles varied as a function of total WMH volume. No significant interactions between timepoint and distance were found, confirming that the spatial patterns did not significantly change over the two-year period.

3.1.3 Longitudinal microstructural changes in WMH penumbra

Independent of the spatial gradients, FWF and MDt exhibited a significant interaction between WMH load and timepoint ($p = 0.018$ and 0.012 for FWF and MDt, respectively), with subjects with higher WMH loads showing greater longitudinal increases in these metrics. Significant main effects of WMH load were observed for FWF and all tissue diffusivity metrics ($p = 0.011$ for RDt and all $p < 0.001$ for FWF, ADt, and MDt), indicating an overall increase in these metrics in subjects with higher WMH burden. Although subtle, there was evidence for overall longitudinal increases in FWF, MDt, and RDt (all $p < 0.01$).

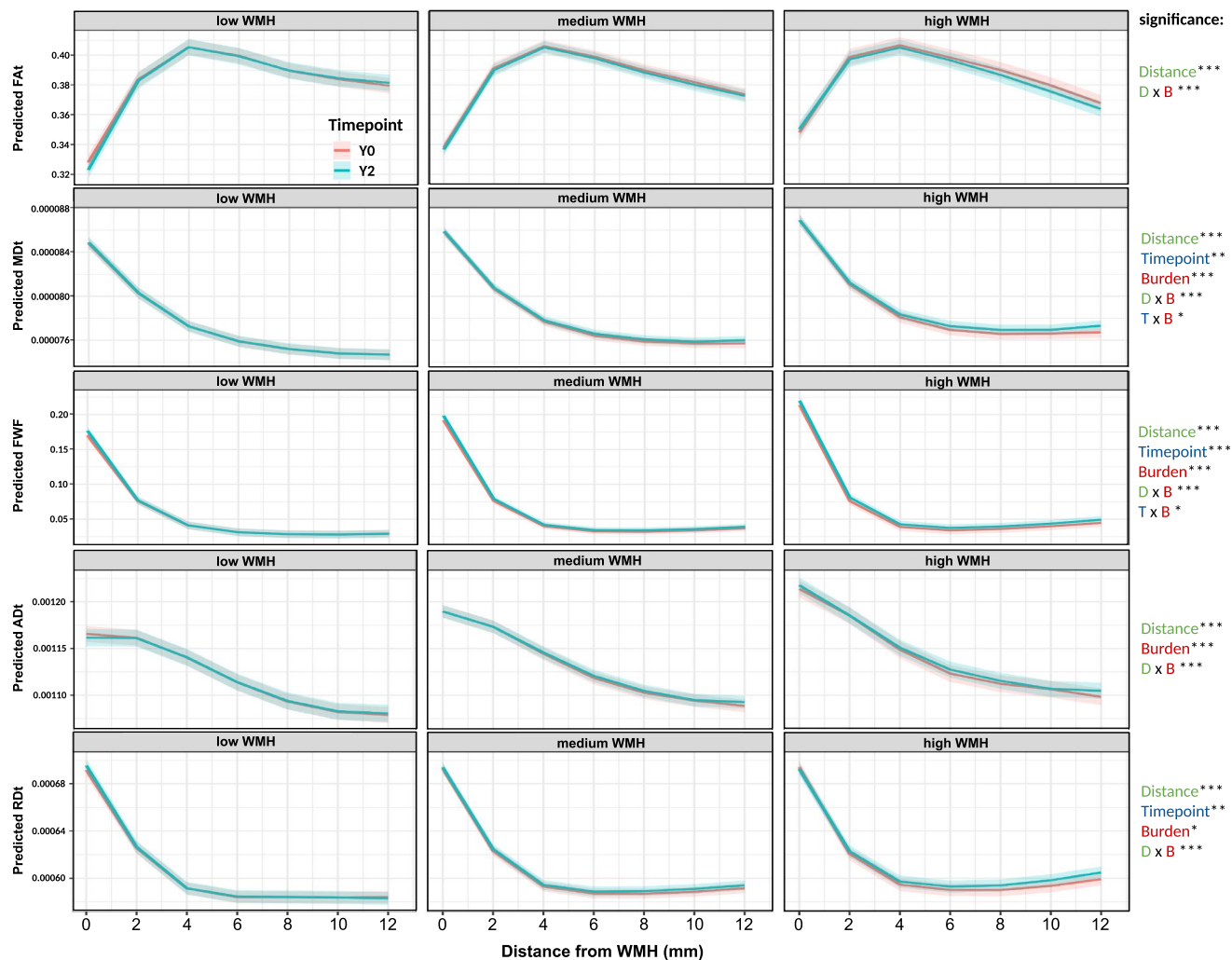


Figure 4: Linear mixed models of diffusion metrics across distance, timepoint, and WMH load.

Separate linear mixed models were fitted for each microstructural metric: Free Water Fraction (FWF), tissue-corrected Fractional Anisotropy (FAt), tissue-corrected Mean Diffusivity (MDt), tissue-corrected Axial Diffusivity (ADt), and tissue-corrected Radial Diffusivity (RDt). Each model included fixed effects for spatial distance (Distance), time (Timepoint), global WMH load (Burden), and their interactions (D x B, T x B, D x T), with age and sex as covariates. Subject-specific random intercepts accounted for repeated measures across layers and timepoints. Spatial distance from the WMH boundary (0–12 mm) was modeled using a 6th-degree orthogonal polynomial to capture non-linear microstructural gradients. Although the global WMH load is a continuous variable, we stratified it into three tertial groups of low, medium and high WMH burden to visualize the microstructural gradients at different WMH burden. On the right panel, the significant main effects and interactions are indicated for each metric, with asterisks denoting statistical significance (* $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$).

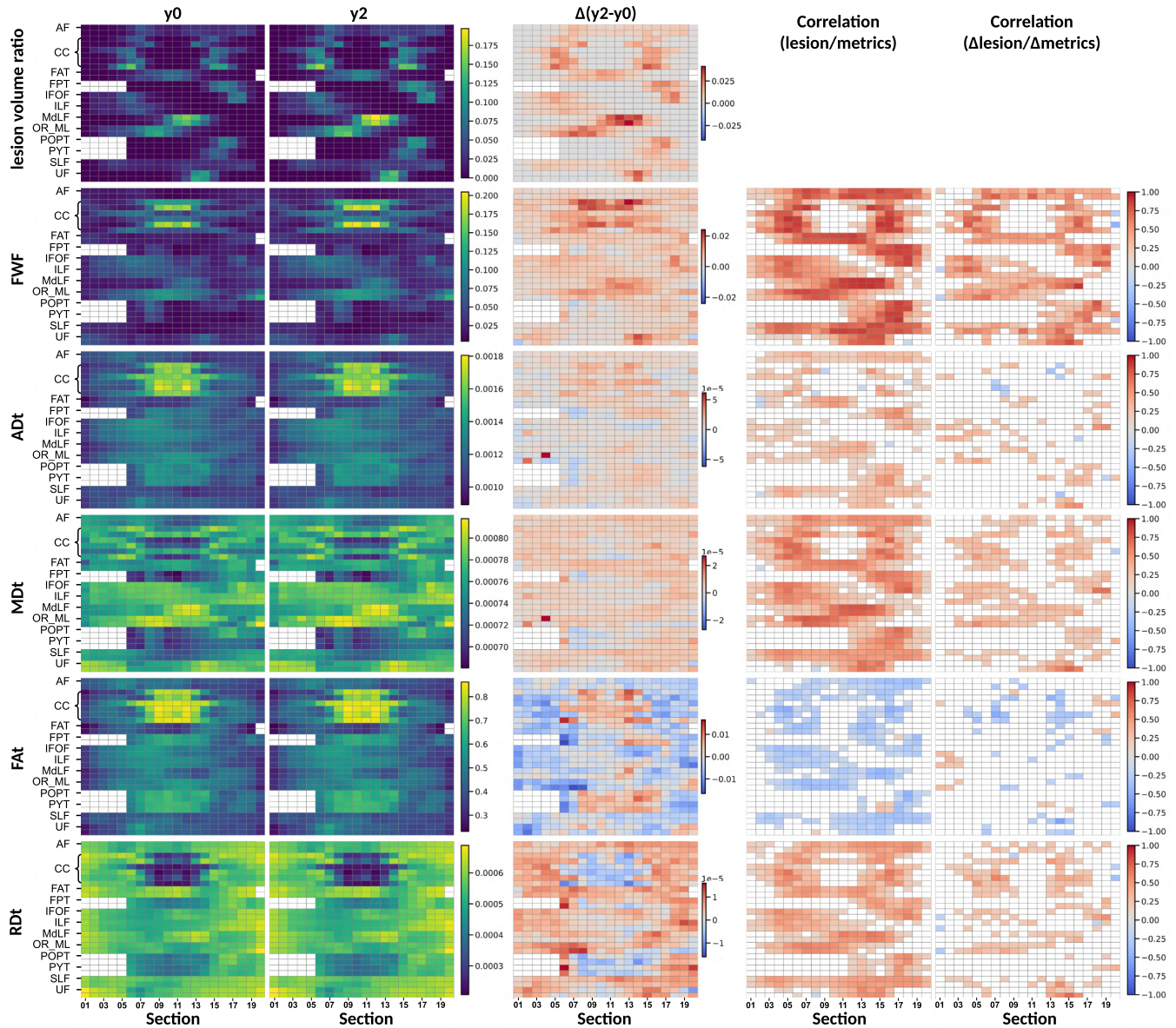


Figure 5: Population-level bundle profiles

Population-level profiles of the different bundles (from top to bottom: AF, arcuate fasciculus; CC, corpus callosum (Fr_1 and Fr_2: frontal, Oc: occipital, Pa: parietal, Pr_Po: precentral–postcentral, Te: temporal); FAT, frontal aslant tract; FPT, frontopontine tract; IFOF, inferior fronto-occipital fasciculus; ILF, inferior longitudinal fasciculus; MdLF, middle longitudinal fasciculus; OR_ML, optic radiation; POPT, parieto-occipital pontine tract; PYT, pyramidal tract; SLF, superior longitudinal fasciculus; UF, uncinate fasciculus) are shown for the left (top row) and right (bottom row) hemispheres across multiple metrics, together with the mean inter-acquisition differences at the individual level. WMH volume ratio was calculated as the volume of the WMH relative to the segment volume for each individual and each segment. Additionally, correlations between the WMH volume ratio and other metrics were computed for each segment, as well as correlations between the longitudinal changes in WMH volume ratio and the corresponding changes in the other metrics. All correlation values were corrected for multiple comparisons using the Benjamini–Hochberg false discovery rate (FDR) correction. Sections 1 to 5 of the FPT, POPT, and PYT bundles were excluded due to poor tracking at the pontine level. Similarly, section 20 of the FAT bundle was removed because it exhibited tracking defects in the majority of subjects.

3.2 Tractometry

3.2.1 Overview of the bundle profile

Figure 5 y0 and y2 columns provide the overview of population-level bundle profiles. The top row highlight specific bundle segments that are more prone to WMH. Bundles in close contact with periventricular regions exhibit the highest lesion-to-bundle volume ratios. The peripheral sections of the CC are particularly affected, along with long association pathways such as the IFOF, MdLF, and SLF. In contrast, projection bundles that traverse the periventricular space more briefly show more localized WMH involvement, including the FPT, POPT, and PYT. Bundle profiles of diffusion metrics show overall spatial patterns of each metric along the bundles. We note the elevated FWF in the central sections of the CC, likely reflecting partial-volume effects from the ventricles due to the thinness of this region. Regarding longitudinal changes in bundle profiles (third column), ADt and MDt did not exhibit substantial temporal variations across bundles, nor did they show distinct spatial patterns. In contrast, FAt and RDt displayed more heterogeneous and opposing changes over time, while FWF generally showing an increase. In the CC, the central region became more anisotropic, accompanied by FWF increases, whereas peripheral regions exhibited increased isotropy. Similar patterns were observed in the PYT and POPT in sections adjacent to the ventricles (segments 10-15), with decreased FAt and increased RDt, indicating a reduction in anisotropy and a loosening of fiber organization within these segments. Despite these consistent trends, the overall magnitude of longitudinal changes across diffusion metrics remained modest relative to their absolute values.

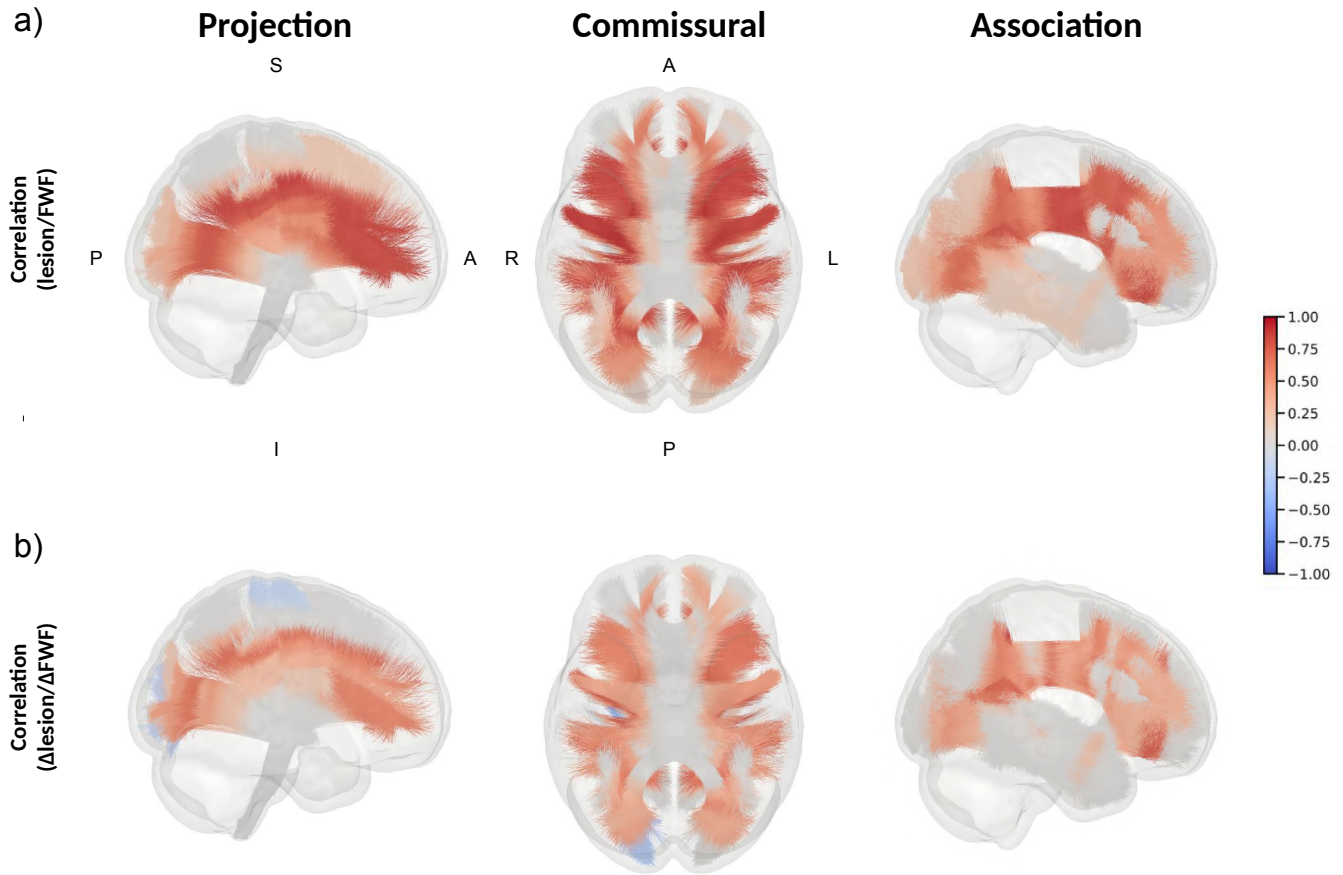


Figure 6: Visualization of both inter-individual and longitudinal correlations between FWF and WMH volume.

(a) Correlation maps between FWF and WMH volume ratio across white matter bundle segments. Correlations were computed within each segment across all bundles. Bundles are grouped into three categories: projection tracts (right sagittal view: PYT, POPT, OR_ML, FPT), commissural tracts (inferior view: all CC bundles), and association tracts (right sagittal view: AF, FAT, IFOF, ILF, MdLF, SLF, UF). Orientations: P, posterior; A, anterior; S, superior; I, inferior; R, right; L, left. (b) Correlation maps between changes in FWF and changes in WMH volume ratio from Y0 to Y2 across white matter bundle segments.

3.2.2 WMH burden effects on white matter bundle profiles

Correlation analyses between the segment-wise WMH volume fraction and diffusion metrics (Fig.5, fourth column) revealed a strong and wide-spread positive association between FWF and WMH volume fraction (Fig.6.a), with a marked accumulation of FWF across all segments containing lesions. For tissue-specific tensor metrics, both MDt and RDt showed strong positive correlations with higher WMH volume fraction across lesioned segments. In contrast, FAt exhibited a consistent negative correlation in lesioned regions, whereas ADt demonstrated a weaker and less consistent association across bundle segments. Longitudinal analyses (Fig.5, fifth column) showed that increases in FWF within lesioned segments were strongly correlated with segment-specific WMH volume progression over time (Fig.6.b). Among diffusion metrics,

only MDt displayed a global positive longitudinal association at the bundle level, although the magnitude of this relationship remained modest. The remaining diffusion metrics exhibited weaker and more spatially heterogeneous correlations across segments.

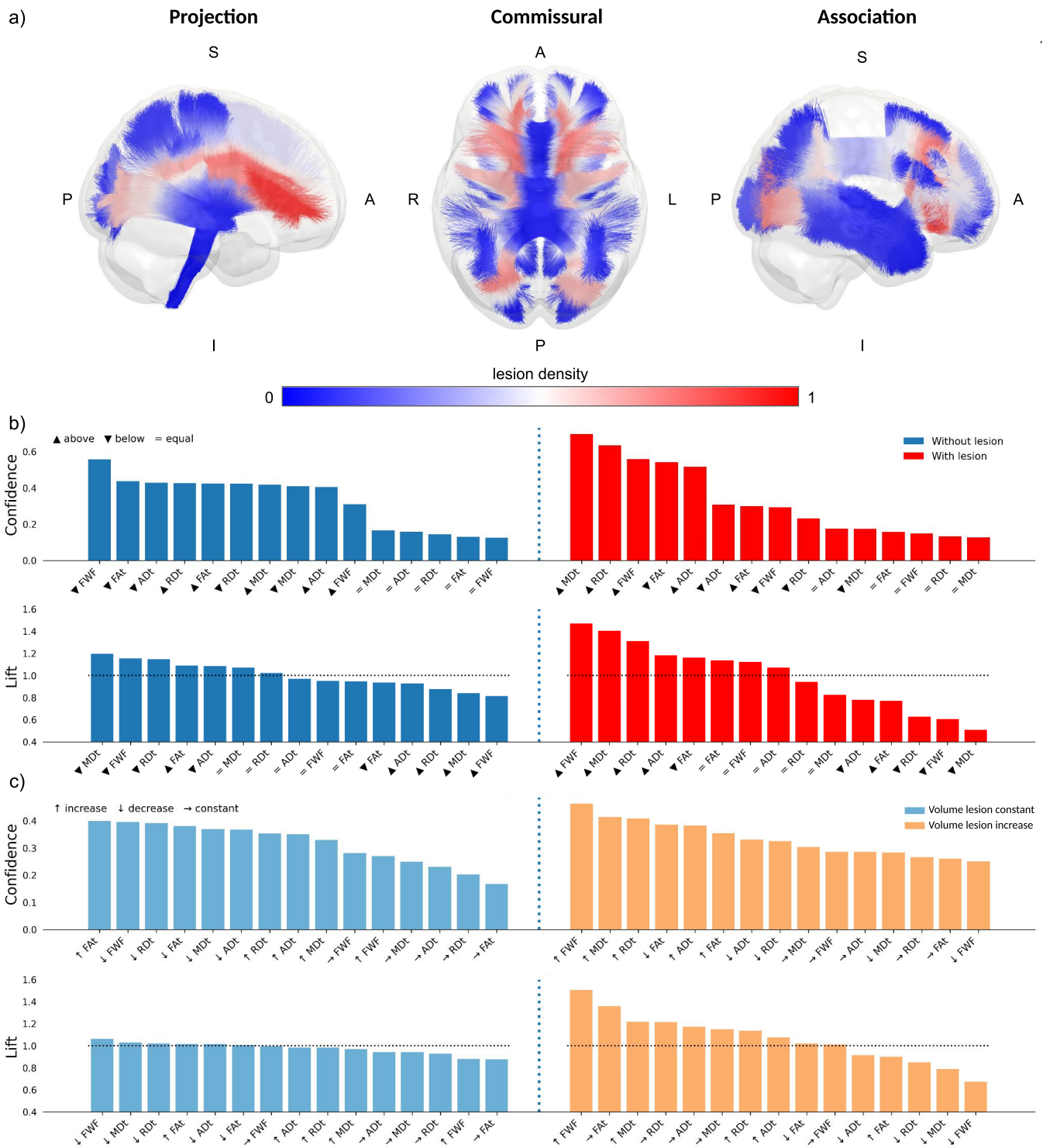


Figure 7: Association analysis of DTI metric changes with WMH presence and volume using lift and confidence.

(a) WMH density maps across bundle segments. WMH density was computed within each segment across all white matter bundles. Bundles were classified into three major categories: projection tracts (right sagittal view), commissural tracts (inferior view), and association tracts (right sagittal view). Orientations are indicated as follows: P, posterior; A, anterior; S, superior; I, inferior; R, right; L, left. (b) Confidence and lift values were calculated to assess the inter-individual associations between DTI metric states and WMH status (with vs. without lesion). Metric values were classified as above (▲), below (▼), or equal (=) to reference values derived from lesion-free segments. Confidence quantifies how frequently a given diffusion metric state occurs among segments presenting a WMH (or no lesion), with values ranging from 0 to 1. Lift evaluates whether the co-occurrence between WMH status and a given metric state is greater (lift > 1), lower (lift < 1), or comparable (lift ≈ 1, indicated by the black dashed line) to what would be expected by chance. (c) Association metrics were similarly computed for longitudinal analyses, examining the relationships between changes in DTI metrics and WMH volume evolution. Metric changes were categorized as increases (↑), decreases (↓), or constant (→) values relative to reference values. Confidence reflects the probability of observing a specific metric change given a pattern of WMH evolution, while lift quantifies the strength of this association compared with chance expectations.

3.3 Data-mining

3.3.1 Population-level effects

In tractometry data, 71.6% of bundle segments were lesion-free, whereas 28.4% contained WMHs in at least one subject. Consistent with Fig.3.a, the segments most affected by WMHs were located in the periventricular regions, particularly around the superior and inferior horns of the lateral ventricles (Fig.7.a). The FP Growth data mining of the tractometry-based diffusion profiles in association with WMH presence (Fig.7.b) revealed several associations of moderate strength as measured by lift ($L > 1.2$), in contrast to associations in lesion-free segments that showed only weak associations ($L < 1.2$). The weak associations in non-lesional segments likely represent imperfectly symmetrical discretization due to slightly skewed distribution of the diffusion metrics around the mean. Most prominent diffusion profiles associated with WMH included *MDt above* ($L = 1.404$, $C = 0.70$), *RDt above* ($L = 1.310$, $C = 0.64$), and *FWF above* ($L = 1.470$, $C = 0.56$). For the FAt and ADt metrics, although high confidence was observed for *FAt below* ($L = 0.54$) and *ADt above* ($L = 0.52$), the corresponding lift values remained low ($C = 1.16$ and $C = 1.18$, respectively), indicating that these metric patterns did not co-occur with WMH presence more frequently than expected by chance.

3.3.2 Longitudinal changes across diffusion metrics

In the longitudinal dataset, bundle sections with stable WMH volume showed only very weak associations between WMH status and diffusion changes, with confidence values ranging from 0.17 to 0.40 and lift values close to 1.0, indicating little specificity or consistent associations when segment-specific WMH is stable (Fig.7.c). Bundle sections with an increase in WMH volume showed a strong association with FWF increase, with a confidence of 0.463 and a lift of 1.509, and a moderate association with *MDt increase* ($L=1.271$, $C=0.414$). Other associations showed relatively high confidence values, such as *RDt increase* ($C = 0.41$), *FAt decrease* ($C = 0.39$), and *ADt increase* ($C = 0.38$), but their corresponding lift values indicated very weak associations. WMH volume increase showed a strong association with *FAt constant*, suggesting limited coupling between WMH volume progression and changes in FA values.

4 Discussion

4.1 lesion-gradient-based analyses

4.1.1 Spatial distribution of WMH and microstructural implications

The regions most consistently affected by WMHs are located around the lateral ventricles, particularly adjacent to the temporal (posterior) and frontal (anterior) horns. Despite this periventricular concentration, the overall distribution remains diffuse across large portions of the white matter, with many voxels affected only in a small subset of individuals. Such variability poses a challenge for anatomical analyses because each individual presents a distinct pattern of WMH distribution and, consequently, a different profile of white matter degradation. These observations align with prior neuroimaging studies describing a predominant distribution of WMHs around the frontal and occipital horns of the lateral ventricles (DeCarli et al., 2005; Frey et al., 2019; Lee et al., 2009; Thammasart et al., 2025). Such a pattern underscores the particular vulnerability of periventricular projection and commissural fibers, which are frequently intersected by WMHs and show microstructural disruption measurable with DTI (Maillard et al., 2011; O'Sullivan et al., 2001; Taylor et al., 2017). Tractography-based analyses further support these findings, demonstrating that WMH burden is closely tied to altered integrity in periventricular tracts but has less impact on long-range fibers situated in the posterior fossa (Alber et al., 2019; Liu et al., 2021). Collectively, this evidence highlights the importance of WMH topography in understanding the heterogeneous impact of WMHs on brain connectivity, pointing to a particular susceptibility of tracts traversing periventricular regions. The spatial organization of WMHs supports the notion that a whole-bundle approach may not be the most informative, as WMH localization suggests focal alterations within the cerebral microstructure. A segment-wise, tractometry-style approach is therefore more informative for capturing these localized changes.

4.1.2 Perilesional white matter alterations beyond visible lesions

Microstructural alterations in perilesional NAWM follow clear and robust spatial gradients extending up to 12 mm from the WMH border. Across all metrics, WMH tissue exhibits the strongest abnormalities, which gradually normalize with increasing distance, reflecting a chronic state of disrupted white matter organization (Maillard et al., 2014; Promjunyakul et al., 2015). Beyond these spatial effects, subtle longitudinal increases in FWF and MDt may reflect slow, diffuse progression of extracellular and tissue-related alterations. Together, these findings reinforce the concept of a structured WMH penumbra,

where microstructural changes extend beyond the visible hyperintensity and continues to evolve subtly over time (Kern et al., 2023; Mayer et al., 2022). However, the different diffusion metrics are not affected uniformly: FAt and ADt appear less sensitive than MDt, FWF, and RDt, [a difference that may reflect demyelination, axonal rarefaction, and increased extracellular water \(Silbert et al., 2024\)](#) [although DTI metrics cannot directly confirm these specific tissue substrates without histological validation](#). Our results extend this knowledge by [describing](#) not only the presence of such alterations but also their systematic spatial organization across diffusion metrics, as well as the dependence of these alterations on the overall WMH burden. These findings underscore the limitations of binary [WMH](#) maps, which capture only overt damage while overlooking broader perilesional changes (Rahmani et al., 2024). The combination of stable diffusion gradients and a sharp FWF drop at the [WMH](#) edge highlights the potential of advanced microstructural imaging markers to better characterize perilesional NAWM and improve assessment of white matter pathology contributing to functional decline. The increased FWF in periventricular white matter may result from impaired integrity of the ependymal barrier at the level of the lateral ventricles, facilitating cerebrospinal fluid leakage into the surrounding tissue, [although this represents a plausible biological mechanism rather than a directly demonstrated mechanism](#). These results emphasize the importance of sensitive diffusion-derived metrics in characterizing the evolution of white matter pathology (Visser et al., 2021). In particular, FWF provides a powerful marker of dynamic tissue change, capable of capturing [microstructural changes that may](#) reflect pathological processes beyond the visible [WMH](#) borders. This is especially relevant in aging and small vessel disease, where conventional [WMH](#) load may underestimate the extent of microstructural damage (Mayer et al., 2022). Moreover, the ability of the FW-DTI model to separate extracellular fluid content from white matter fiber integrity may be particularly valuable in the context of clinical trials targeting axonal or myelin repair. The perilesional penumbra approach presented here would be critical for the monitoring of therapeutic effects that extend beyond established WMH themselves.

4.2 Tractometry

4.2.1 FWF along white matter bundle profiles

The cross-sectional and longitudinal correlation analyses of tractometry data [provide additional, convergent information](#) on diffusion-derived alterations associated with focal white matter lesions. Consistent with the lesion-gradient analyses, FWF emerges as the most sensitive marker of lesion-related changes in the tractometry-based analyses as well, showing a strong and consistent positive association with [WMH](#) volume both across segments and over time. This pronounced accumulation of free water within lesioned

regions suggests an expansion of the extracellular fluid content abnormalities that evolves in parallel with WMH progression. In periventricular regions, particularly within the corpus callosum, the marked increase in FWF observed in the presence of WMHs may be compatible with infiltration of cerebrospinal fluid into the surrounding white matter (Chang et al., 2023). This process may result from reduced impermeability of structural barriers at the WMH site, facilitating the diffusion of CSF into the parenchyma (Visser et al., 2021), potentially contributing to local loss of tissue integrity, though this interpretation remains hypothetical. While partial-volume effects related to ventricular proximity, especially in the central part of the CC (sections 8–14), contribute to the overall spatial distribution of FWF, the robustness of its association with WMH burden suggests that FWF variations are not solely attributable to geometric contamination, although a residual partial-volume contribution cannot be fully excluded.

4.2.2 DTI metrics along white matter bundle profiles

Changes observed in diffusion tensor-derived metrics are broadly compatible with patterns of reduced microstructural integrity associated with white matter lesions, though DTI metrics remain indirect measures whose relationship to specific tissue changes cannot be confirmed without histological data. The mild increase in MDt and RDt, together with the reduction in FAt within lesioned segments, but with inconsistent ADt suggests relatively mild secondary tissue microstructural alterations, such as myelination disruption or inflammation are plausible candidates suggested by the broader literature, but cannot be specifically confirmed from these metrics alone. The relative preservation of ADt may tentatively argue against pronounced axonal loss, although this inference should be interpreted cautiously given the non-specificity of DTI-derived measures.

4.3 Data-mining

4.3.1 Key microstructural alterations underlying WMH

We applied the FP-Growth data-mining approach on tractometry data for its efficiency in handling high-dimensional data and its ability to uncover co-occurring patterns across metrics and bundles, in order to identify frequent, interpretable associations between WMH status and diffusion alterations at the bundle-section level. Consistent with the tractometry-based correlation analyses, both WMH presence and progression are primarily linked to increases in MDt, RDt, and FWF, pointing primarily to expansion of the extracellular space, possibly with mild loss of myelin integrity, without strong evidence for axonal degeneration. Notably, the strong association between WMH volume increase and FWF emphasizes the

accumulation of free water as a key marker of lesion-related extracellular changes, complementing the correlational observations described above.

In contrast, FAt and ADt showed only weak or minimal associations with WMH presence or progression, reinforcing the notion that axonal integrity is largely preserved, and that FAt, even after FW correction, is not particularly sensitive marker to track WMH-related microstructural changes.

Overall, the FP-Growth results offer preliminary insights that appear consistent with the tractometry findings, suggesting a pattern of lesion-driven tissue alteration characterized by extracellular water expansion and microstructural rarefaction. Given that this data-mining framework has not previously been applied to tractometry data and that findings derive from a single cohort without external validation, these observations should be interpreted as exploratory rather than confirmatory. Nonetheless, they provide an encouraging basis for further investigation into RDt and MDt as potentially more specific and interpretable markers than FAt for tracking WMH evolution in CSVD, particularly when combined with bundle-profile analyses.

4.4 Limitations and Future Directions

A primary limitation of this work lies in the heterogeneity of acquisition protocols within the ADNI3 dataset, particularly the coexistence of single-shell and multi-shell diffusion MRI data in the “basic” and “advanced” protocols. After internally assessing the comparability of FW-DTI estimates derived from these acquisitions, we adopted a pragmatic harmonization strategy based on single-shell FW-DTI metrics. However, recent work has demonstrated that estimating the bi-tensor free-water model from single-shell data is intrinsically under-constrained and may introduce systematic bias in estimates of free-water fraction and free-water-corrected diffusion metrics (Golub et al 2021, Correia et al 2024 PMID 40800265). Although these findings were reported for a regularized gradient-descent implementation rather than the AMICO framework used here, the underlying limitation reflects the information content of single-shell acquisitions and may therefore affect any single-shell implementation. Consequently, the allocation of the diffusion signal into free-water and tissue compartments is likely to be less robust than with multi-shell acquisitions, and changes attributed to FWF and tissue-specific metrics (MDt and RDt) should be interpreted with caution. Nevertheless, the observed tract-segment-specific increases in FWF, MDt, and RDt are consistent with the known and expected neuropathology of WMH, reflecting increased extracellular free water alongside tissue microstructural degeneration. Thus, while the relative contributions of these compartments should be confirmed in the future investigations using fully optimised multi-shell protocols, the spatially coherent pattern of associations support the biological relevance of

the reported findings.

Another shared limitation across approaches concerns the challenge of defining a truly lesion-free control baseline. WMH are highly prevalent in aging, even among cognitively normal individuals, making it difficult to establish an unaffected reference. Within the data-mining framework, alternative definitions of “lesion-free” baselines could be explored. For example, participants without WMHs not only in a specific tract segment but along the entire bundle (or at least within a predefined spatial distance) could be selected to minimize the influence of perilesional microstructural abnormalities, as highlighted in the lesion-gradient analysis. Furthermore, because this approach relies on discretizing metric properties (e.g., “above” or “below” baseline), future studies could introduce more refined categories to capture graded deviations from normative values.

Beyond population- and acquisition-related variability, the analysis of NAWM depends directly on the segmentation method employed. For a given lesion, different tools (such as BIANCA (Griffanti et al., 2016), LST (Guo et al., 2019), or DeepMedic (Kamnitsas et al., 2017)) may yield variable WMH core volumes. These discrepancies can in turn shift the definition of NAWM regions of interest and influence downstream interpretations. In the present study, we selected SHIVAI due to its demonstrated sensitivity and precision, as well as its suitability for large-scale applications (Tsuchida et al., 2024).

An important limitation of our approach concerns the potential confounding between distance to WMHs and distance to the lateral ventricles. In our cohort, as in most studies of age- or vascular risk-related WMHs, WMHs are preferentially concentrated in periventricular regions (see Sections 3.1.1 and 4.1.2), such that distance to a WMH and distance to the ventricles are strongly spatially correlated. It is well established that FWF can be artificially elevated near the ventricles due to partial volume effects with cerebrospinal fluid (CSF) [ref. to be added, e.g. Pasternak et al., 2009; Maillard et al., 2013], independent of any local lesional pathology. Since our current model does not include ventricular distance as a covariate, we cannot exclude the possibility that the observed FWF gradient near WMHs reflects, at least in part, a partial volume artifact related to ventricular proximity rather than a genuine perilesional increase in extracellular water. This interpretation is consistent with our finding of a particularly marked effect in the periventricular corpus callosum (Section 4.2.1), a region anatomically adjacent to the ventricles. Future studies incorporating ventricular distance as an independent covariate, or employing dedicated partial volume correction approaches, will be needed to disentangle the respective contributions of lesional pathology and ventricular proximity to the observed FWF variations.

A further limitation concerns the fact that the lesion-gradient analysis pools all WMHs together, irrespective of anatomical location, presumed origin, or size. Regarding size, a stratification would in principle be relevant if lesions were isolated and independent of one another; however, in a cohort with high

lesion density, small WMHs located close to larger lesion volumes are influenced by the latter, both in the lesion itself and in the surrounding NAWM, so this neighborhood contamination makes a size-based dissociation unreliable in the present context. Regarding presumed origin (vascular versus neurodegenerative), our choice to focus on a cognitively normal population without neurodegenerative disease was intended to favor WMHs of predominantly vascular origin, a distinction documented in the literature (Parent et al., 2026). Regarding anatomical location, a periventricular/deep white matter classification was not adopted, as the definition of deep white matter is not consensual across studies, confluent lesions spanning both compartments would require an arbitrary split, and prior work has reported no significant difference in tissue impact according to lesion location in a vascular context (Schmidt et al., OHBM 2026, abstract). We further acknowledge that the diffusion metrics used here remain susceptible to local white matter anatomy independently of WMH presence. An ideal correction would rely on a voxel-wise comparison against a lesion-free cohort; however, using a young, lesion-free reference would introduce an age-related bias, whereas an older reference cohort would leave some white matter regions with too few unaffected voxels to provide a reliable baseline, since these regions are rarely spared by WMHs in older populations. Given these constraints, we chose to pool lesions according to location rather than stratify by size or origin, while acknowledging that this choice introduces additional variability in the reported values.

Regarding tractometry, several limitations are intrinsic to the methodology. The analyzed bundles were generated using the BundleSeg atlas and therefore represent standardized white matter pathways. Alternative frameworks (such as XTRACT (Warrington et al., 2020), TractSeg (Wasserthal et al., 2018), or WMQL (Wassermann et al., 2016)) could be employed in future work to vary both the number of bundles analyzed and their anatomical definitions.

As the data-mining strategy has not previously been applied to tractometry, several methodological refinements are possible. The current classification scheme relies on a baseline derived from a subset of the studied cohort, which may limit generalizability. Moreover, for interpretability, metrics were grouped into three categories. Introducing greater granularity could better reflect varying degrees of deviation from baseline values and enhance sensitivity to subtle microstructural alterations. Furthermore, it should be emphasized that the FP-Growth-derived marker selection reported here was conducted in a single cohort without validation in an independent dataset. As such, the identification of FWF, RDt and MDt as the most informative diffusion metrics for characterizing WMH-related microstructural alterations should be regarded as hypothesis-generating rather than conclusive. Replication in an external cohort will be necessary before these findings can support interpretive frameworks or inform clinical decision-making.

Finally, the present work identified subtle yet significant microstructural changes associated with WMH

progression over a two-year period, detectable through both conventional analyses and the data-mining framework. Future longitudinal studies with extended follow-up durations (≥ 6 years) would be valuable for capturing clinically meaningful progression and for better characterizing the impact of newly emerging WMHs (Schmidt et al., 2012; van Leijsen et al., 2018). In addition to FW-DTI modeling, other multi-compartment diffusion approaches such as NODDI (Zhang et al., 2012) may provide complementary insights into WMH-related tissue alterations. Beyond free-water contamination, the interpretability of tensor-based metrics is further limited by the presence of complex and crossing fibre populations, a well-known constraint of the diffusion tensor model (Descoteaux et al., 2007). While fODF reconstruction in our pipeline already relies on constrained spherical deconvolution (CSD) for tractography, the diffusion metrics analyzed here remain tensor-derived. Future work could therefore benefit from complementary CSD-based approaches, such as tissue-specific relative signal fractions (Khan et al., 2021; Mito et al., 2020) or fixel-based fibre density and cross-section measures (Raffelt et al., 2017), which are less confounded by fibre crossing and could offer a more anatomically specific characterization of WMH-related microstructural changes. Moreover, the data-mining framework could be applied at multiple spatial scales to identify the most informative microstructural features and white matter bundles, thereby reducing the arbitrariness of metric and bundle selection in future analyses.

5 Conclusion

Our analyses suggest that white matter WMHs are predominantly associated with increased free water, potentially related to compromised ependymal barrier integrity at periventricular WMH sites, and more subtle microstructural alterations within the surrounding penumbra. Among diffusion metrics, FWF emerges as a sensitive marker of WMH progression, while MDt and RDt capture increases in water diffusivity perpendicular to fibers, consistent with reduced structural barriers and possible loss of myelin integrity. These observations, initially established through tractometry and NAWM analyses, are further echoed by the exploratory FP-Growth findings, which, despite their preliminary nature, point to a convergent direction. Across the analyses, relying on FAt alone, even with FW correction, appears insufficient, and RDt and MDt seem better suited for characterizing WMH-related microstructural alterations. Together, these findings support the potential value of multi-metric diffusion analyses for understanding lesion-induced white matter damage and its evolution over time, while calling for replication in independent cohorts before stronger conclusions can be drawn.

Data and Code Availability

Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (<https://adni.loni.usc.edu>). The ADNI dataset is publicly available to qualified investigators upon registration and completion of a data use agreement.

The following open-source pipelines and software were used in this study:

- **TractoFlow**: <https://github.com/scilus/tractoflow>
- **Tractometry_Flow**: https://github.com/scilus/tractometry_flow
- **RBx_Flow**: https://github.com/scilus/rbx_flow
- **FreeWater_Flow**: https://github.com/scilus/freewater_flow
- **Scilpy library**: <https://github.com/scilus/scilpy>
- **SHIVAI** ([WMH](#) segmentation): <https://github.com/pboutinaud/shivai>

All custom scripts developed for statistical analyses, lesion-gradient modeling, and data-mining procedures are openly available at https://github.com/ThoumyreStanislas/statistical_analyses.

Author Contributions

Stanislas Thoumyre: Writing – Original draft, Data curation, Visualization, Validation, Methodology, Formal analysis, Conceptualization. **Ami Tsuchida**: Writing – Validation, Methodology, Formal analysis, Review, Editing. **Arnaud Boré**: Data curation. **Maxime Descoteaux**: Validation, Review, Editing. **Marc Joliot**: Validation, Review, Editing. **Laurent Petit**: Conceptualization, Validation, Review, Editing. **François Rheault**: Conceptualization, Methodology, Validation, Review, Editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary Data for: Comprehensive diffusion MRI analysis reveals key microstructural alterations underlying white matter hyperintensities

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1 Tractoflow configuration details

All subjects were processed following the pipeline illustrated in Figure S1. The scripts used throughout the pipeline are described in the TractoFlow documentation for preprocessing steps up to tractogram generation, and in the respective GitHub repositories for FreeWater_flow, BundleSeg, and Tractometry_flow. Fiber reconstruction was performed using a local tracking algorithm implemented on GPU through the scilpy scripts. The entire pipeline was executed using the container scilus_1.6.0.sif.

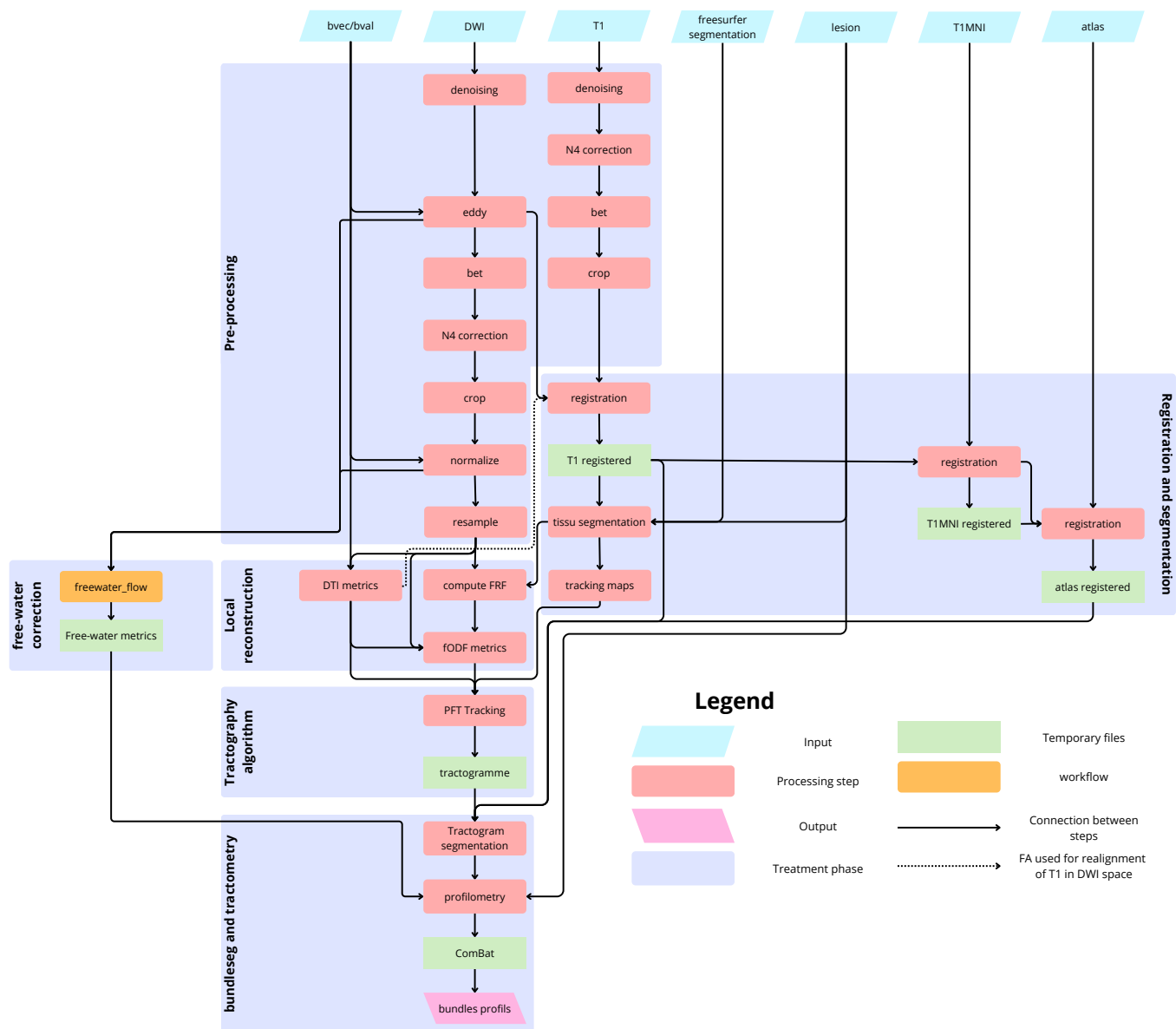


Figure 1: Processing pipeline up to the generation of bundle profiles

2 The impact of tissue segmentation method

Some parameters were adjusted to better accommodate an aging population. In particular, the ABS profile was used for the generation of tracking and seeding maps (Figure S2). Indeed, atlas-based segmentation approaches, such as recon-all-clinical implemented in FreeSurfer, are preferable to intensity-based segmentation methods (e.g., FAST) in the presence of white matter lesions, as the latter may lead to tissue misclassification. In addition, lesion masks were incorporated to ensure comprehensive inclusion of white matter tissue during tractography.

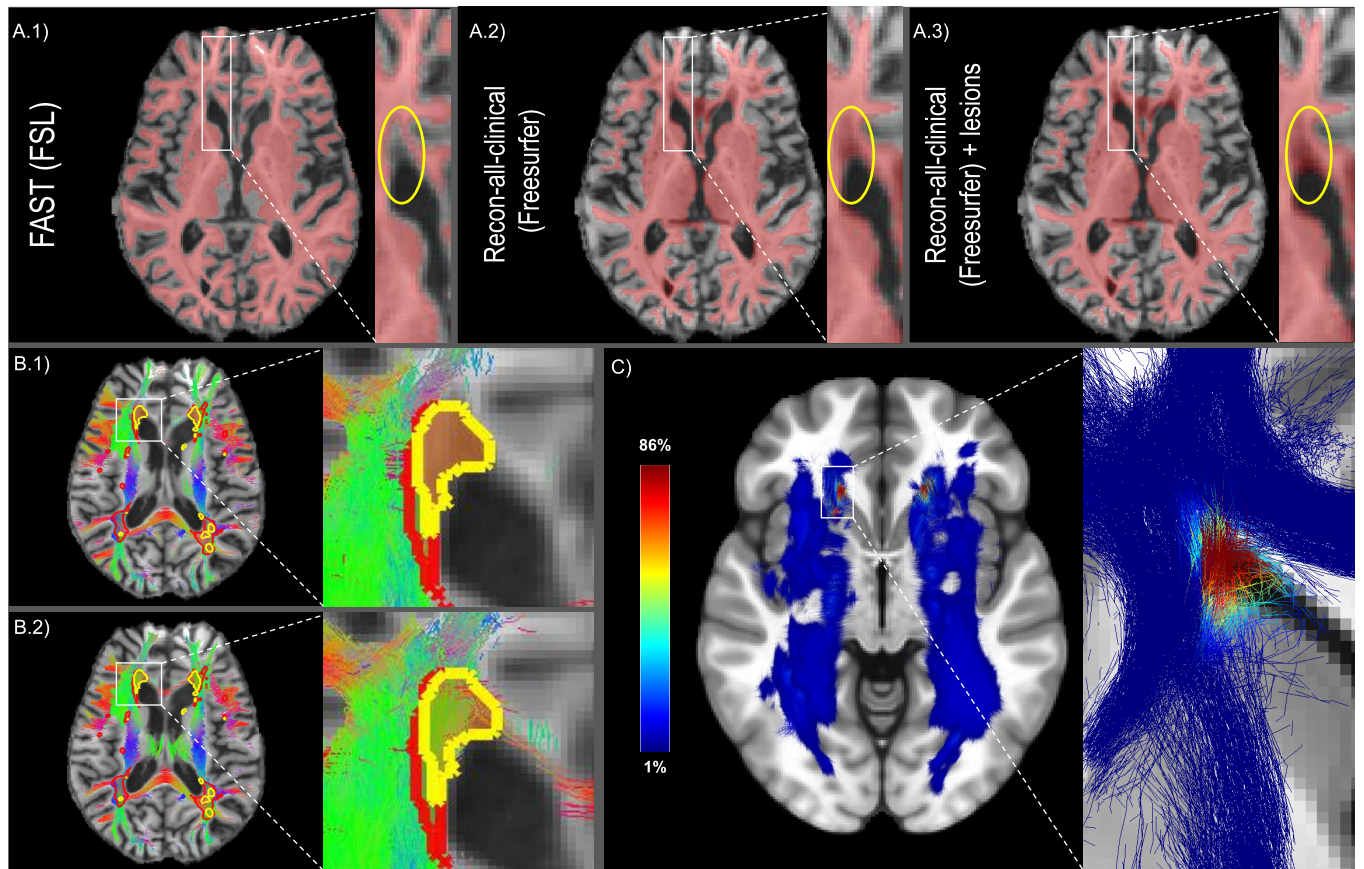


Figure 2: Impact of white matter segmentation methods on the tractogram

(A) Visualisation of the white matter mask A.1) Segmented by FAST, A.2) Segmented by recon-all-clinical and A.3) Segmented by recon-all-clinical plus correction by the lesion mask. The yellow circle shows a periventricular lesion. **(B)** Effect of mask correction on local tracking. Red delineation corresponds to WMHs segmented by SHIVA-WMH, yellow delineation corresponds to the portion of the lesion not taken into account by FAST white matter segmentation and corrected by Recon-all-clinical. B.1) Local tracking in the case of FAST segmentation B.2) Local tracking in the case of recon-all-clinical segmentation plus lesion mask correction. **(C)** Population average map showing where streamlines are not correctly reconstructed during FAST segmentation. The higher the percentage, the greater the number of individuals with reconstruction defects in this zone. The set of streamlines comes from a concatenation of the different bundles of each individual obtained with recon-all-clinical plus lesions.

3 Single vs multi-shell FW comparison

Another important aspect of our study concerns the multisite nature of ADNI, which results in protocol variability across subjects and longitudinal acquisitions, particularly with respect to diffusion imaging. Two acquisition schemes were present in the dataset: single-shell (SS) and multi-shell (MS). A given subject could have a single-shell acquisition at baseline and a multi-shell acquisition at follow-up, or exclusively SS or MS across time points. This protocol heterogeneity introduces a potential bias in free-water estimates, as differences in acquisition schemes may induce systematic shifts in free-water-corrected diffusion metrics. Consequently, diffusion measures derived from SS and MS acquisitions cannot be directly compared without additional harmonization. We therefore investigated three complementary questions: first, whether single-shell data could be reliably used for free-water correction; second, whether free-water estimates derived from SS acquisitions were correlated with those obtained from MS data; and third, whether the acquisition-related shift could be mitigated by simulating single-shell data from multi-shell acquisitions, thereby enabling comparisons across the full dataset.

To assess whether single-shell (SS) data can be reliably used for free-water correction, and whether SS-derived estimates are consistent with multi-shell (MS) estimates, we performed comparative analyses across several external datasets (UK Biobank; Cambridge Centre for Ageing and Neuroscience; Human Connectome Project) in addition to ADNI (Figure S3). For these comparisons, SS data were simulated from MS acquisitions by retaining only diffusion-weighted volumes with b -values ≤ 1200 s/mm². Within ADNI, we specifically selected subjects who had SS acquisitions at baseline and MS acquisitions at follow-up. Free-water fraction (FWF) values were compared within corresponding white matter bundles. Statistical comparisons were conducted using mixed-effects linear regression models applied to mean FWF values, accounting for repeated measures where appropriate. For the remaining datasets, diffusion MRI acquisition parameters were as follows. The UK Biobank (UKB) dataset was acquired with a repetition time (TR) of 3600 ms, 50 diffusion-weighted directions per shell, b -values of 0, 1000, and 2000 s/mm², and a multiband acceleration factor (MB) of 3. The Cambridge Centre for Ageing and Neuroscience (CamCAN) dataset was acquired with a TR of 9100 ms, 30 diffusion-weighted directions, b -values of 0, 1000, and 2000 s/mm², and MB = 3. Finally, the Human Connectome Project (HCP) data were collected with a TR of 5520 ms, 90 diffusion-weighted directions, b -values of 0, 1000, 2000, and 3000 s/mm², and MB = 3.

With the exception of the HCP, which includes very high diffusion shells $b = 3000$ s/mm², the remaining datasets showed strong mixed-model coefficients and consistent associations between SS- and MS-derived FWF estimates. These results support the feasibility of using SS data for free-water correction and for subsequent analyses across heterogeneous acquisition protocols.

Finally, using the analysis illustrated in Figure S4, we evaluated whether converting multi-shell (MS) acquisitions to single-shell estimated (SSe) could harmonize free-water-corrected diffusion metrics with the single-shell data. The first observation is that MS-derived metrics differ significantly from SS-derived metrics, with an increase in FA-t and a decrease in MD-t. The second observation is that after converting MS data to SSe, FA-t values no longer show significant differences compared to SS data. Regarding MD-t, only the Frontal Aslant Tract and the right Superior Longitudinal Fasciculus bundles exhibited significant differences between SSe and SS data.

However, it is important to note that there is a two-year interval between the MS and SS acquisitions for ADNI data. The differences observed between the metrics in Figure 4, as well as the coefficient values reported in Figure 3, may partly reflect changes due to the time elapsed. Moreover, this comparison is limited by the absence of test-retest data in the ADNI population, which are unfortunately unavailable.

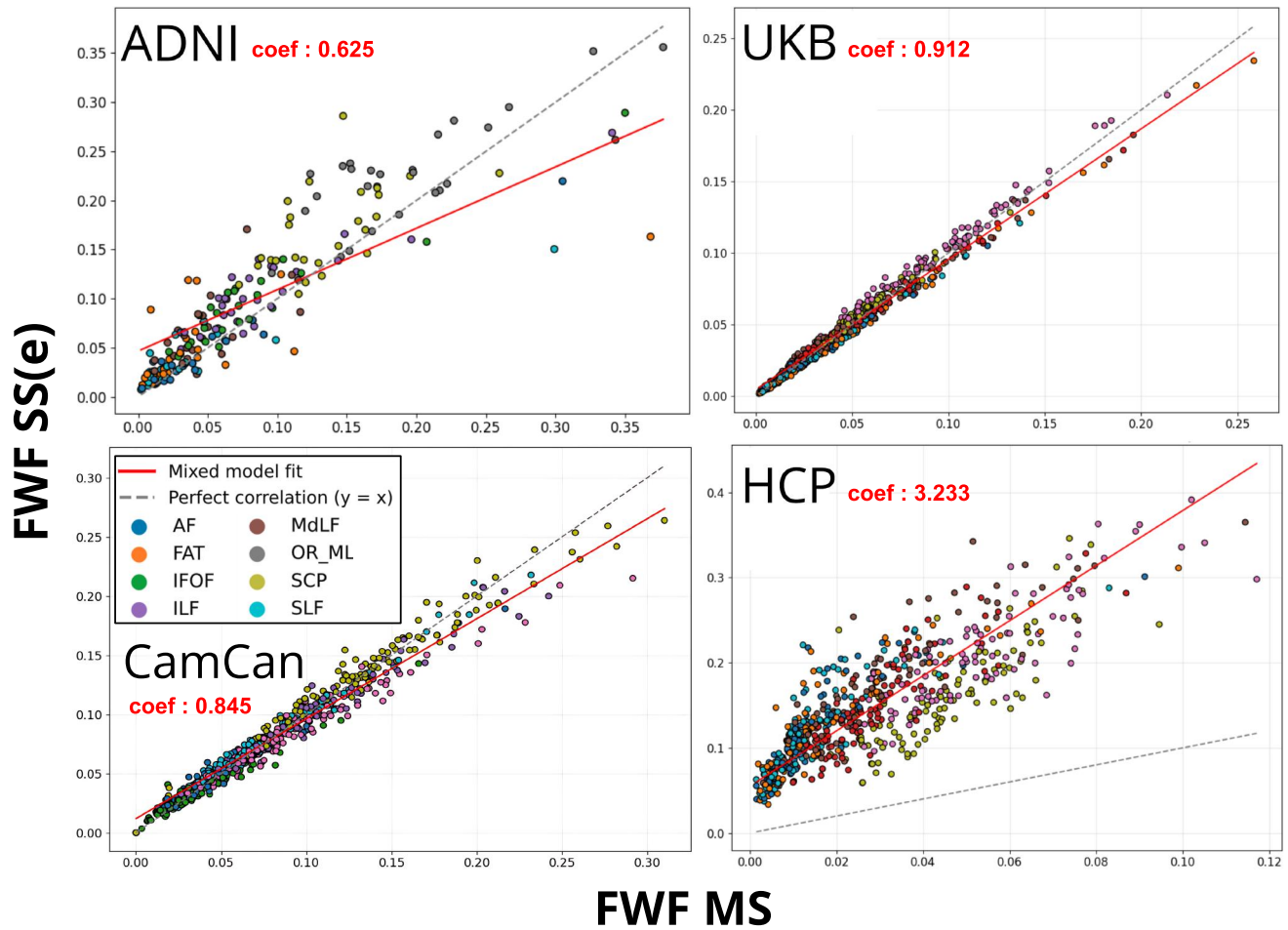


Figure 3: Mixed-effects linear regression of mean free-water fraction (FWF) values between multi-shell (MS) and single-shell (SS). Comparison of FWF values between MS and SS for ADNI and single-shell estimated (SSe) from multi-shell for the rest of dataset, across seven white matter bundles: AF (Arcuate Fasciculus), FAT (Frontal Aslant Tract), IFOF (Inferior Fronto-Occipital Fasciculus), MdLF (Middle Longitudinal Fasciculus), OR ML (Optic Radiation Meyer's Loop), SCP (Superior Cerebellar Peduncle), and SLF (Superior Longitudinal Fasciculus).

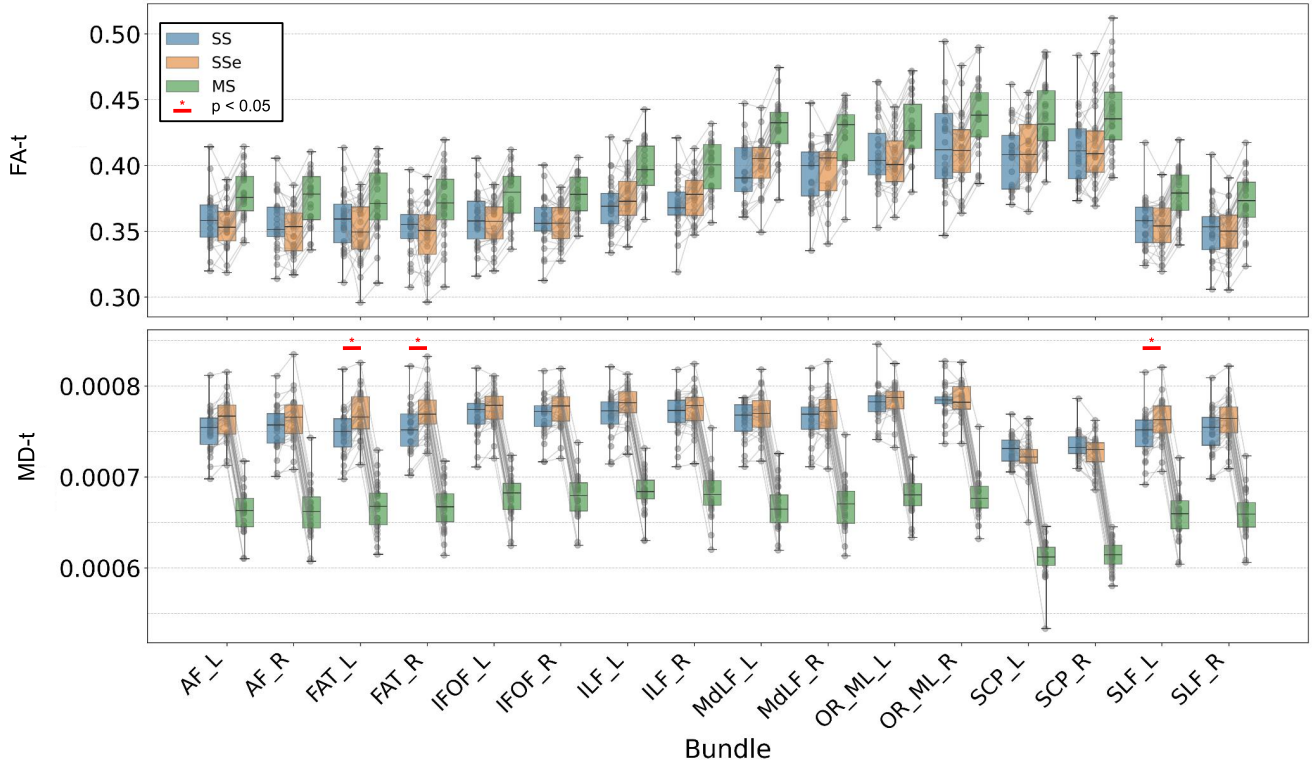


Figure 4: **Comparison of Free Water metrics between single-shell (SS), Single-shell estimated (SSe), and multi-shell (MS) data.** Comparison of free-water corrected metrics (FA-t and MD-t) from ADNI data ($n=28$) between SS, SSe, and MS acquisitions, across bundles: AF (Arcuate Fasciculus), FAT (Frontal Aslant Tract), IFOF (Inferior Fronto-Occipital Fasciculus), MdLF (Middle Longitudinal Fasciculus), OR ML (Optic Radiation Meyer's Loop), SCP (Superior Cerebellar Peduncle), and SLF (Superior Longitudinal Fasciculus).

4 voxel-wise distribution

Prior to averaging, the voxel-wise distribution of each diffusion metric within each ROI was inspected using the D'Agostino-Pearson omnibus normality test, together with skewness and excess kurtosis. The D'Agostino-Pearson test, which combines skewness and kurtosis to assess overall departure from normality, returned $p < 0.001$ across all ROIs. However, given the large number of voxels per ROI (ranging from 10,648 to 46,837), this test is highly sensitive and detects trivial deviations from normality that are not necessarily of practical relevance; it was therefore complemented by effect-size indicators. Skewness, which quantifies the asymmetry of the distribution around its mean (0 = symmetric, positive = right-shifted tail), ranged from 0.242 to 0.457 across ROIs, reflecting mild but consistent positive skewness. Excess kurtosis, which compares tail weight to that of a normal distribution (0 = Gaussian-like,

positive = heavier tails, negative = lighter tails), ranged from -0.478 to 0.216 , remaining close to 0. Overall, although the D'Agostino-Pearson test indicated a statistically significant departure from normality (expected given the sample size), skewness and excess kurtosis showed only modest and consistent deviations across ROIs, supporting the use of the mean as a representative summary statistic per ROI.

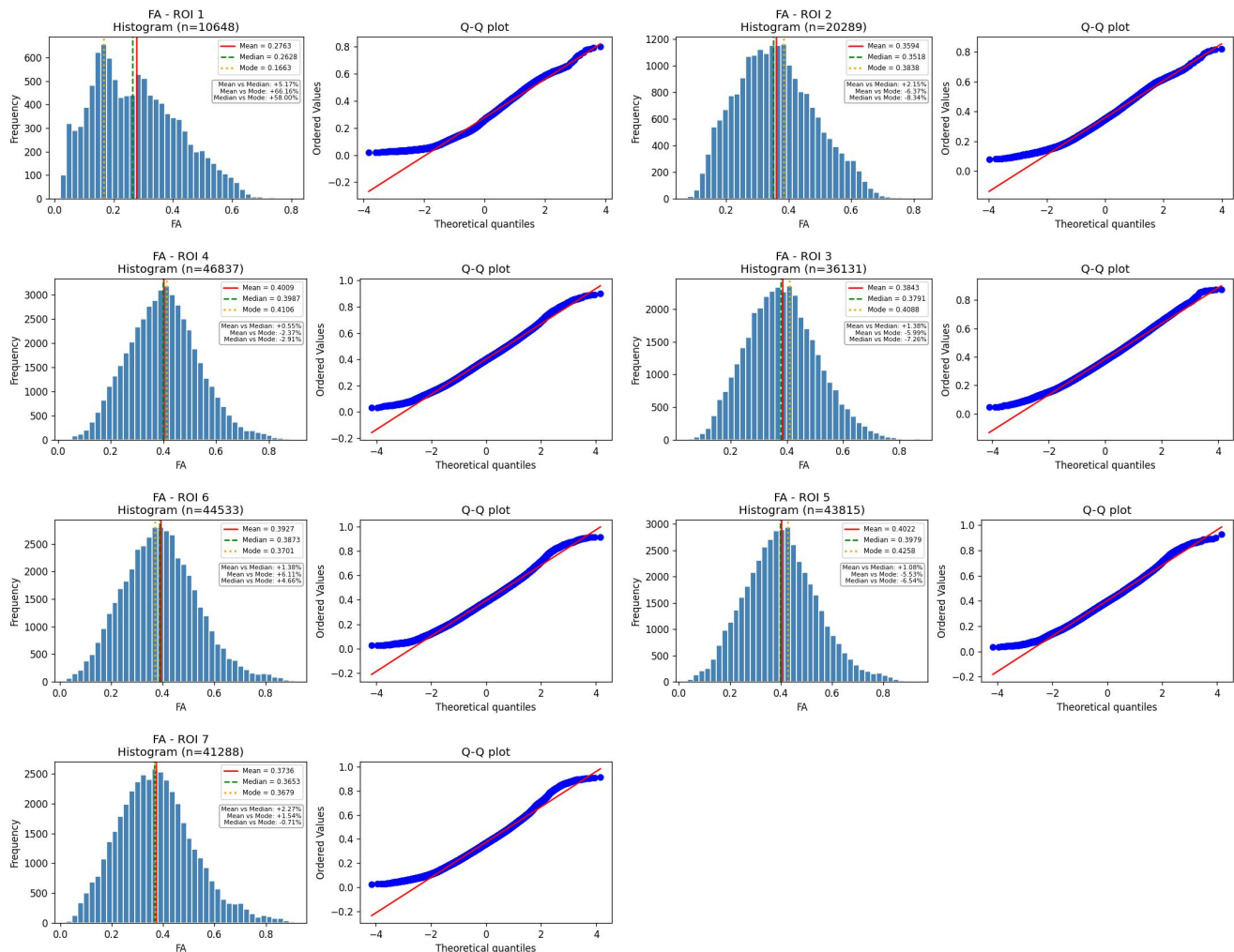


Figure 5: Voxel-wise distribution of FA values within each NAWM ROI (ROI 1–7, panels A–G). For each ROI, the left panel shows the histogram of voxel-wise FA values (40 bins) with the mean (red solid line), median (green dashed line), and mode (orange dotted line; center of the most populated histogram bin) indicated, along with their pairwise percentage differences. The right panel shows the corresponding Q-Q plot against a theoretical normal distribution. n indicates the number of NAWM voxels included per ROI after exclusion of NaN and zero values.

Note. Excess kurtosis close to 0 indicates tail weight comparable to a Gaussian distribution; positive skewness indicates a right-shifted tail. n = number of NAWM voxels in the ROI after exclusion of NaN and zero values.

5 Visual presentation of bundles studied

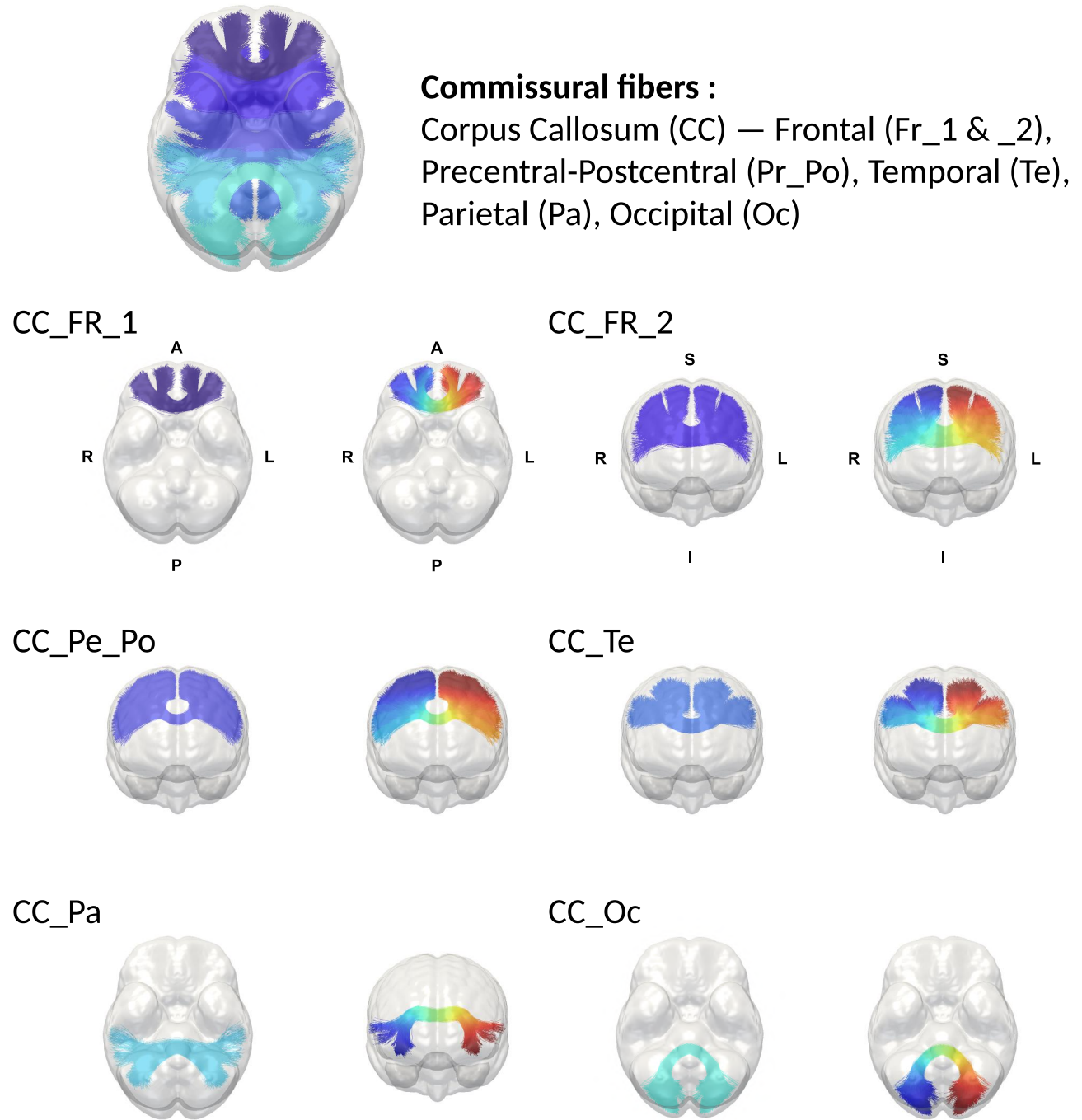


Figure 6: Presentation of the commissural bundles.

Overview of white matter bundles following segmentation with BundleSeg and their tractometry-derived profiles. Sections are color-coded from blue (section 1) to red (section 20). Anatomical orientation is indicated by A (anterior), P (posterior), S (superior), I (inferior), L (left), and R (right).

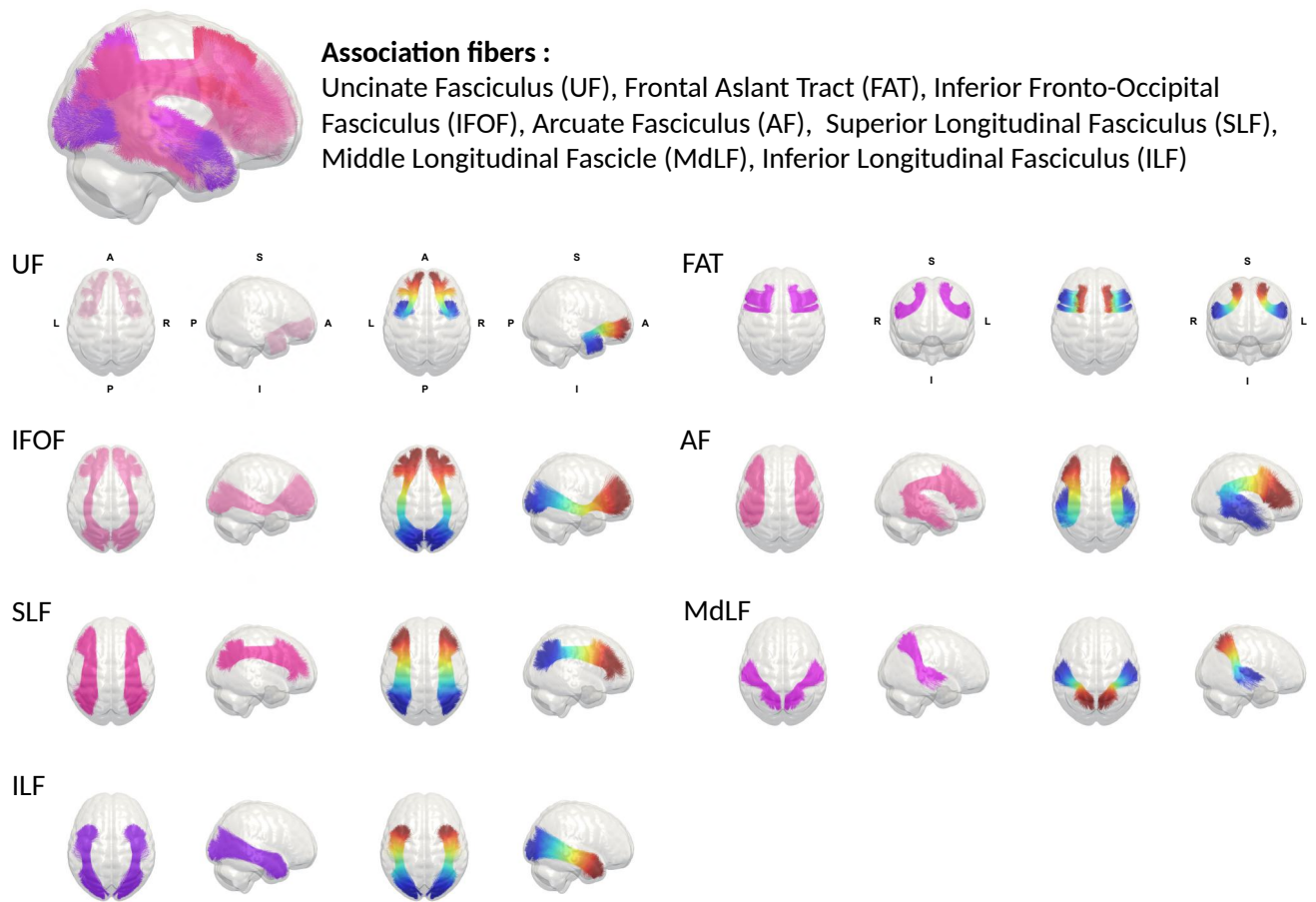


Figure 7: Presentation of the association bundles.

Overview of white matter bundles following segmentation with BundleSeg and their tractometry-derived profiles. Sections are color-coded from blue (section 1) to red (section 20). Anatomical orientation is indicated by A (anterior), P (posterior), S (superior), I (inferior), L (left), and R (right). Only the right-sided bundle is displayed in the right sagittal view.

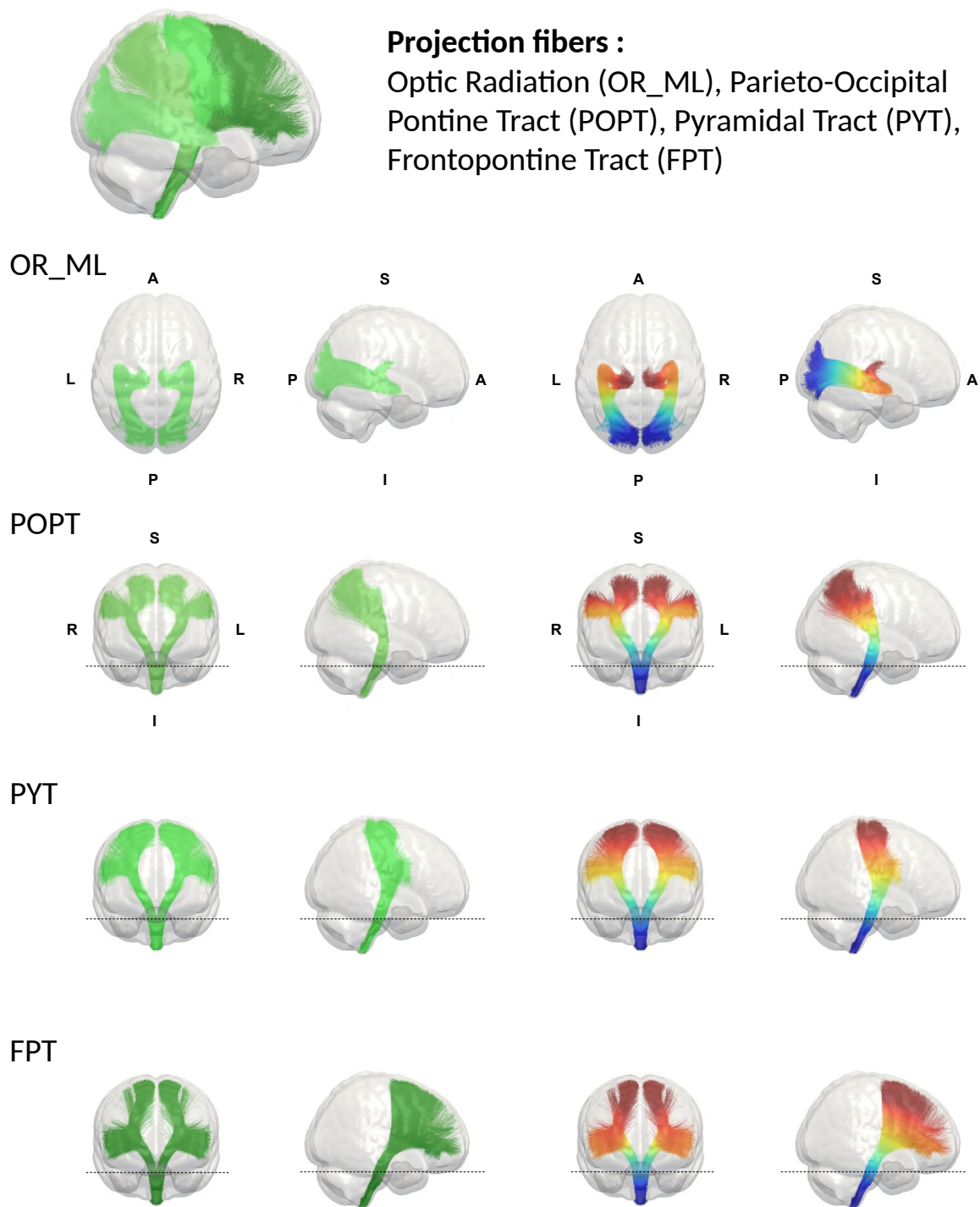


Figure 8: Presentation of the projection bundles.

Overview of white matter bundles following segmentation with BundleSeg and their tractometry-derived profiles. Sections are color-coded from blue (section 1) to red (section 20). Anatomical orientation is indicated by A (anterior), P (posterior), S (superior), I (inferior), L (left), and R (right). The black dashed line indicates the segments that were excluded from the tractometry and data-mining analyses for the PYT, POPT, and FPT bundles, as tractography was unreliable in these regions. Only the right-sided bundle is displayed in the right sagittal view.

6 Practical recommendations for standard DTI datasets

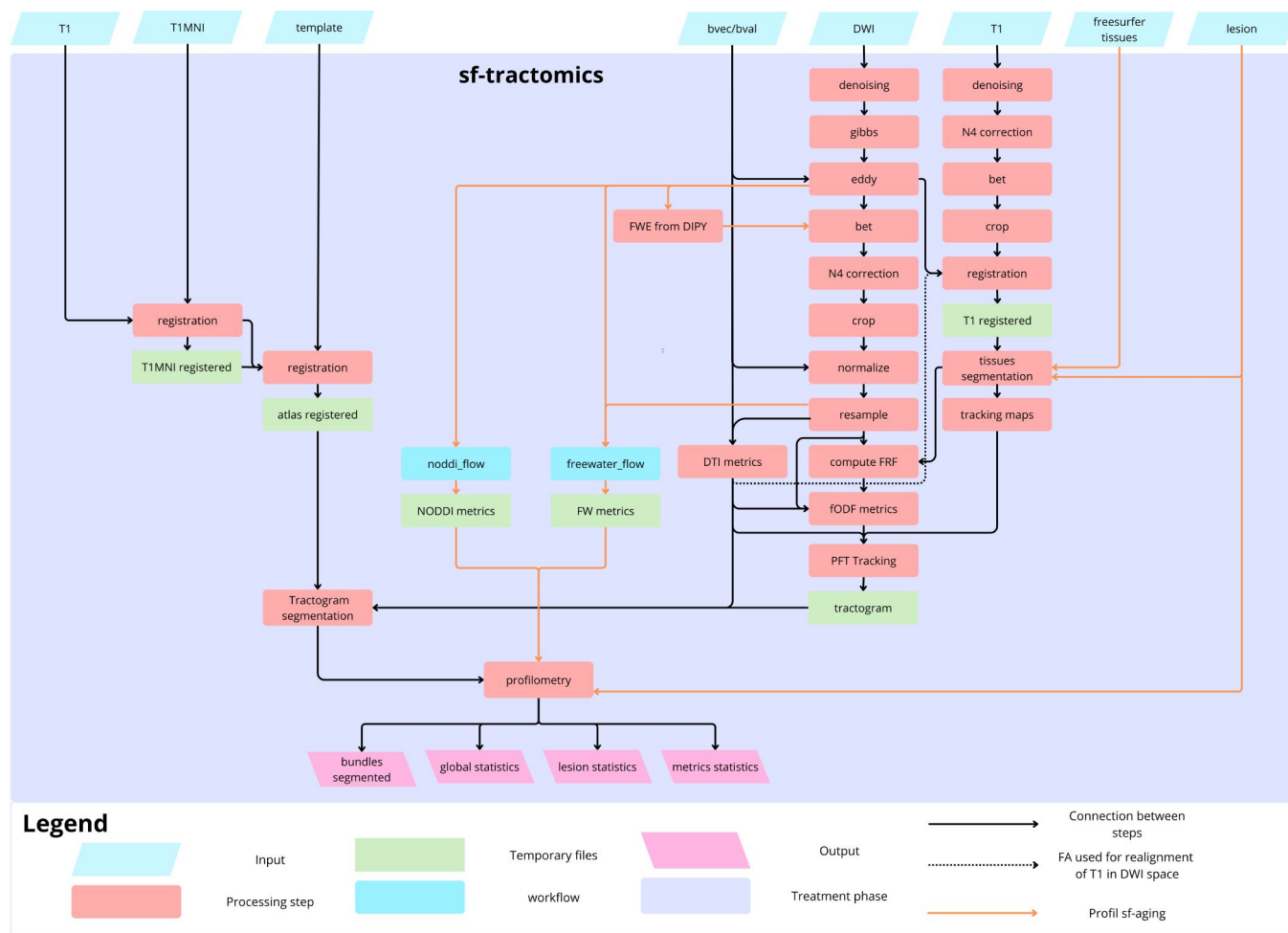


Figure 9: Overview of the recently released sf-tractomatics pipeline and its dedicated sf-aging profile (orange arrows), designed for diffusion MRI processing in aging populations. Pink boxes denote the common processing steps of the pipeline (denoising, eddy-current correction, registration, particle-filtering tractography), light-blue boxes denote the dedicated sub-workflows (noddi_flow, freewater_flow), and green boxes denote intermediate files. The sf-aging profile introduces a free-water correction step (FWE from DIPY) upstream of Fiber Response Function estimation, together with the incorporation of lesion masks into tissue segmentation, seeding/tracking, and tractometry.

The sf-aging pipeline is intended to be integrated into the sf-tractomatics ecosystem, whose infrastructure has just been made publicly available, as a dedicated profile accompanied by modules specifically adapted to the processing of images from aging populations. This extension accounts for the neuroanatomical par-

ticularities and microstructural alterations commonly observed in brain aging, in particular the increased burden of white matter (WM) lesions and the associated changes in diffusion properties.

Following the recommendations of (Golub2021), who highlight the limitations of free-water fraction estimation from single-shell diffusion data, the preprocessing stage introduces a dedicated free-water estimation step using the DIPY library. This step yields a diffusion image corrected for the free-water contribution in WM, which is then fed back into the pipeline upstream of Fiber Response Function (FRF) estimation, improving tensor modeling and the subsequent reconstruction of fiber orientations.

A second key adaptation concerns the explicit handling of WM lesions during tractography: lesion masks are incorporated into the seeding and tracking maps to limit segmentation errors and spurious bundle reconstructions arising from tissue alterations. Anatomical segmentation likewise relies on atlas-based approaches rather than purely intensity-based methods, using FreeSurfer's recon-all to obtain a robust parcellation consistent with cortical and subcortical structures, as opposed to intensity-driven tissue models such as FAST.

When diffusion data allow it, sf-aging further supports more advanced microstructural modeling: for multi-shell acquisitions, the NODDI model is estimated, providing ODI and NDI metrics that offer a finer characterization of tissue microstructure and can be incorporated into tractometry analyses. Lesion maps are also propagated into the tractometry stage itself, allowing lesion burden to be quantified along reconstructed WM bundles and enabling direct assessment of its impact on tract-specific microstructural integrity.

Finally, sf-tractomics provides a centralized, MultiQC-based reporting system that allows rapid and systematic quality inspection of every processing step for each subject, confirming that the pipeline ran correctly and that bundles were segmented satisfactorily. Where this is not the case, these reports facilitate the identification of subjects or bundles showing anomalies or segmentation failures, improving the robustness and reliability of cohort-level analyses.